

LieBN: Batch Normalization over Lie Groups

Ziheng Chen, Yue Song, Rui Wang, Xiao-Jun Wu, and Nicu Sebe

Abstract—Manifold-valued measurements are prevalent in various machine learning tasks. Recent advances have extended Deep Neural Networks (DNNs) to operate on manifolds, accompanied by normalization techniques tailored to different geometries, collectively referred to as Riemannian normalization. However, most existing Riemannian normalization methods are either designed for specific manifolds or fail to effectively normalize manifold-valued sample distributions. To address these limitations, we propose LieBN, a framework for Riemannian Batch Normalization (RBN) over Lie groups. Our approach leverages the theoretically convenient left- and right-invariant metrics, which naturally exist in every Lie group, and provides theoretical guarantees for controlling the Riemannian mean and variance. We instantiate LieBN across nine distinct geometries: four on the Symmetric Positive Definite (SPD) manifold, one on the group of rotation matrices, and four on the manifold of full-rank correlation matrices. Notably, among the SPD metrics, we introduce a novel right-invariant metric and extend three existing Lie group structures via matrix power deformation. Extensive experiments on different manifolds validate the effectiveness of our framework. The code is available at <https://github.com/GitZH-Chen/LieBN.git>.

Index Terms—Riemannian batch normalization, Lie groups, symmetric positive definite matrices, rotations, correlation matrices.



1 INTRODUCTION

Over the past decade or so, Deep Neural Networks (DNNs) have achieved significant progress across various scientific fields [4], [5], [6], [7]. Traditionally, DNNs have been developed under the assumption that the latent space of the input data is Euclidean. However, many applications involve non-Euclidean structures, such as manifolds [8]. To address this challenge, researchers have extended various types of DNNs to manifolds, based on the theories of Riemannian geometry [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27].

Motivated by the great success of normalization techniques [28], [29], [30], [31], researchers have sought to devise normalization layers tailored for manifold-valued data. Brooks *et al* [32] introduced Riemannian Batch Normalization (RBN) designed specifically for the Symmetric Positive Definite (SPD) manifold, with the ability to normalize the Riemannian mean. Kobler *et al* [33] extended this approach to further control the Riemannian variance. *However, the above methods are constrained within the Affine-Invariant Metric (AIM) on the SPD manifold, limiting their applicability.* On the other hand, Chakraborty [34] proposed two distinct Riemannian normalization frameworks: one for Riemannian homogeneous spaces [34, Algs. 1-2] and another for matrix Lie groups [34, Algs. 3-4]. *Nonetheless, the normalization designed for Riemannian homogeneous spaces cannot normalize mean nor variance, while the one for matrix Lie groups is confined to a specific type of distance [34, Sec. 3.2].* Meanwhile, Luo [35, Alg. 2] proposed an RBN layer for general geometries. However, similar to [34, Algs. 1-2], it lacks theoretical guarantees for normalizing sample

statistics. Therefore, a principled Riemannian normalization framework capable of controlling both Riemannian mean and variance remains unexplored.

Given that Batch Normalization (BN) [28] serves as the foundational prototype for various types of normalization, our paper focuses on RBN, with the potential to be extended to other normalization variants. Since several manifold-valued measurements form Lie groups, such as Symmetric Positive Definite (SPD) manifolds [3], [36], [37], special orthogonal groups $SO(n)$ [38], and full-rank correlation matrices [3], [39], we further direct our attention to Lie groups. As each Lie group naturally admits left- and right-invariant metrics [40, Ch. 1.2], we propose a principled framework for RBN over Lie groups under invariant metrics, referred to as LieBN. Compared to previous work, our framework provides a theoretical guarantee for normalizing the Riemannian sample mean and variance across general Lie groups.

Empirically, we focus on the SPD, special orthogonal, and full-rank correlation manifolds. On SPD manifolds, we generalize three existing Lie group structures into parameterized ones by matrix power deformation. Additionally, we propose a novel right-invariant metric, which, to the best of our knowledge, is the **first** non-trivial right-invariant SPD metric¹, referred to as Cholesky Right Invariant Metric (CRIM). We then instantiate our LieBN framework on SPD manifolds under these four Lie group structures. For rotation matrices, we adopt the popular bi-invariant metric [38], which will induce two types of LieBN: one w.r.t. left-invariance and another w.r.t. right-invariance. On the correlation manifold, we manifest our LieBN under four recently developed correlation geometries [3], [39]. Besides, we discuss the optimization of the involved correlation-valued parameters. To facilitate usage, we provide a LieBN

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1. Although some metrics are bi-invariant, the associated group structures are commutative [36], [37]. Therefore, their bi-invariance is reduced to left-invariance.

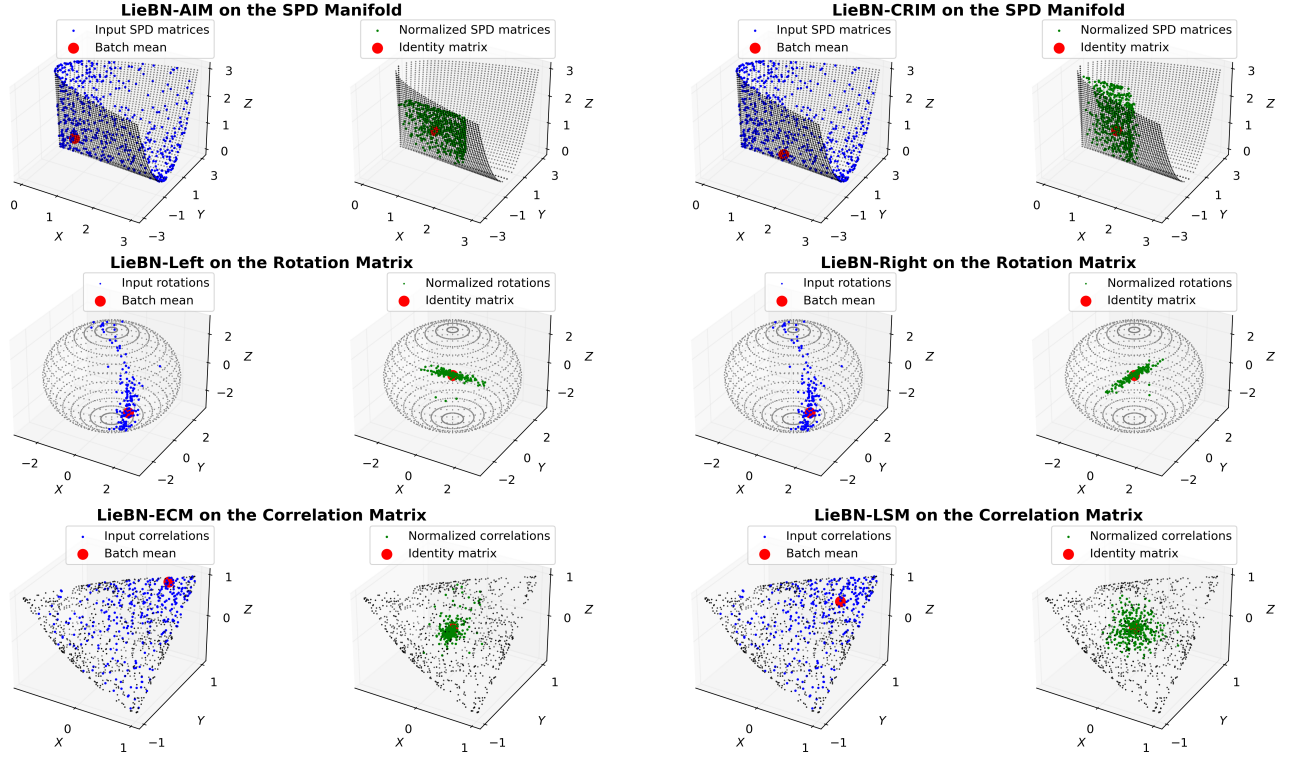


Fig. 1: Illustration of LieBN on the SPD, rotation, and correlation Lie groups. The 2×2 SPD, 3×3 rotation, and 3×3 correlation manifolds can be embedded into $\mathbb{R}^3$ as an open cone [1], a closed ball with antipodal points identified [2], and an open elliptope [3], respectively. LieBN is illustrated by (1) the left-invariant AIM and the proposed right-invariant CRIM geometry on the SPD manifold, (2) left or right translation under a bi-invariant metric on the rotation manifold, and (3) the bi-invariant ECM and LSM geometry on the correlation manifold. On the SPD and correlation manifolds, the batch mean and variance of the same input samples differ under different geometries. In all sub-figures, the black, blue, green, and red dots denote the boundary of the space, the input Lie group samples, the normalized samples, and the batch mean, respectively. As illustrated, our LieBN effectively normalizes the Lie group distribution.

toolbox compatible with PyTorch, which can be used as a drop-in module. Fig. 1 illustrates our LieBN on different geometries, while Fig. 2 illustrate a minimal demo. Extensive experiments on SPD, rotation, and correlation manifolds across three tasks—radar recognition, human action recognition, and electroencephalography (EEG) classification—demonstrate the effectiveness of our methods.

We emphasize that our work is fundamentally distinct from [32], [33], [35] in theory and more general than [34]. Previous RBN methods are either designed for specific geometries [32], [33], [34] or fail to control both the mean and variance [35]. In contrast, our LieBN ensures the normalization of both the mean and variance across general Lie groups. In summary, our main contributions are:

- A general Lie group batch normalization framework with controllable first- and second-order moments;
- A novel right-invariant metric on the SPD manifold, which is the first non-trivial right-invariant SPD metric;
- Concrete instantiations of our LieBN framework on different geometries: four on SPD manifolds, one on the rotation matrices, and four on the correlation manifold;
- Validation of effectiveness of our LieBN framework by extensive experiments on different geometries.

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from LieBN import LieBNSPD, LieBNRot, LieBNCor
from LieBN.Geometry.SPD import SPDMatrices
from LieBN.Geometry.Rotations import RotMatrices
from LieBN.Geometry.Correlation import Correlation

# ==== SPD matrices ====
P_spd = SPDMatrices(n=5).random(4, 2, 5, 5)
# Implemented metrics: LEM, ALEM, LCM, AIM, CRIM
liebn_spd = LieBNSPD([2, 5, 5], metric="LEM",
    batchdim=[0])
output_spd = liebn_spd(P_spd)

# ==== SO(3) matrices ====
P_so3 = RotMatrices().random(4, 2, 3, 3, 3)
# LieBN-Left if is_left else -Right
liebn_so3 = LieBNRot([3, 3, 3], batchdim=[0,
    1], is_left=False)
output_so3 = liebn_so3(P_so3)

# ==== Correlation matrices ====
P_cor = Correlation(n=5).random(4, 2, 5, 5)
# Implemented metrics: ECM, LECM, OLM, LSM
liebn_cor = LieBNCor([2, 5, 5], metric="ECM",
    batchdim=[0])
output_cor = liebn_cor(P_cor)

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Fig. 2: Minimal examples of applying LieBN.

This paper extends our previous conference paper [41] in both theory and implementation. Theoretically, the original LieBN framework was restricted to Lie groups under a left-invariant metric. However, Lie groups also naturally admit right-invariant metrics, which share many theoretical properties with left-invariant ones. Therefore, we generalize LieBN to all natural invariant metrics, including left-, right-, and bi-invariant metrics, providing a more comprehensive framework for RBN. Additionally, we propose a novel non-trivial right-invariant metric on the SPD manifold. In terms of implementation, beyond the original applications to SPD and rotation matrices, we further manifest LieBN on four correlation geometries. Besides, the previous LieBN on $SO(3)$ was based solely on left-invariance and validated on a small dataset. In contrast, this journal submission expands the implementation to both left- and right-invariance and conducts extensive experiments across multiple datasets.

2 PRELIMINARIES

This section briefly reviews Lie groups, as well as the concrete Lie groups of SPD, rotation, and full-rank correlation matrices. For more in-depth discussions, we refer the reader to [42] for smooth manifolds, [40], [43] for Riemannian manifolds, and [44] for Lie groups.

2.1 Lie Groups

Definition 2.1 (Lie Groups [42]). A manifold $\mathcal{M}$ is a Lie group, if it forms a group with a group operation $\odot$ such that $m(\cdot, \cdot) : \mathcal{M} \times \mathcal{M} \ni (x, y) \rightarrow x \odot y \in \mathcal{M}$ and the group inverse $i(\cdot) : \mathcal{M} \ni x \rightarrow x_{\odot}^{-1} \in \mathcal{M}$ are both smooth.

Definition 2.2 (Invariance [40]). A Riemannian metric g^L over a Lie group $\{\mathcal{M}, \odot\}$ is left-invariant, if for any $x, y \in \mathcal{M}$ and $V_1, V_2 \in T_x \mathcal{M}$, it satisfies $g_y^L(V_1, V_2) = g_{L_x(y)}^L(L_{x*,y}(V_1), L_{x*,y}(V_2))$, with $L_x(y) = x \odot y$ as the left translation by x , and $L_{x*,y}$ as the differential map of L_x at y . Similarly, a right-invariant metric g^R satisfies $g_y^R(V_1, V_2) = g_{R_x(y)}^R(R_{x*,y}(V_1), R_{x*,y}(V_2))$, with $R_x(y) = y \odot x$ as the right translation by x , and $R_{x*,y}$ as the differential at y .

A Lie group is both a group and a manifold. The most natural Riemannian metric on a Lie group is the left- or right-invariant metric². A bi-invariant metric has both left- and right-invariance. In this paper, $\{\mathcal{M}, \odot, g\}$, abbreviated as $\mathcal{M}$, always denotes a Lie group with an invariant metric.

The idea of pullback is ubiquitous in differential geometry and can be viewed as a natural counterpart of the bijection.

Definition 2.3 (Pullback Metrics [43]). Suppose $\mathcal{M}_1, \mathcal{M}_2$ are smooth manifolds, g is a Riemannian metric on $\mathcal{M}_2$, and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a diffeomorphism. The pullback of g by f is defined point-wisely as $(f^*g)_p(V, W) = g_{f(p)}(f_{*,p}(V), f_{*,p}(W))$, where $f_{*,p}(\cdot)$ is the differential map of f at $p \in \mathcal{M}_1$, and $V, W \in T_p \mathcal{M}_1$. f^*g is a Riemannian metric on $\mathcal{M}_1$, called the pullback metric of g by f .

Although pullback metrics can also be defined by smooth map [43], we focus on the diffeomorphism. Additionally, if

$\{\mathcal{M}_2, \odot_2\}$ forms a Lie group, the diffeomorphism f can pull back the group operation $\odot_2$ to $\odot_1$ on $\mathcal{M}_1$:

$$P \odot_1 Q = f^{-1}(f(P) \odot_2 f(Q)), \forall P, Q \in \mathcal{M}_1. \quad (1)$$

Definition 2.4 (Weighted Fréchet Mean & Variance [45]). Let $\{w_{1...N}\}$ be weights satisfying a convexity constraint, i.e., $\forall i, w_i > 0$ and $\sum_i w_i = 1$. The weighted Fréchet mean (WFM) of a set of manifold-valued points $\{P_{i...N} \in \mathcal{M}\}$ is defined as

$$\text{WFM}(\{w_i\}, \{P_i\}) = \operatorname{argmin}_{M \in \mathcal{M}} \sum_{i=1}^N w_i d^2(P_i, M), \quad (2)$$

where $d(\cdot, \cdot)$ denotes the geodesic distance. When $w_i = 1/N$ for all i , Eq. (2) is reduced to the Fréchet mean, denoted as $\text{FM}(\{P_i\})$. The Fréchet variance v^2 is the attained value at the minimizer of the Fréchet mean.

On Riemannian manifolds, WFM uniquely exists when samples are locally distributed [46], which are detailed in App. B.1 for completeness. In this paper, we always assume the Fréchet batch mean exists. As we focus on Riemannian manifolds, we will use the terms "Riemannian mean" and "Fréchet mean" interchangeably, as well as "Riemannian variance" and "Fréchet variance."

Basic notations. For Euclidean spaces, we denote $\langle \cdot, \cdot \rangle$ as the canonical inner product over $\mathbb{R}^{n \times n}$, with $\|\cdot\|_F$ as the induced norms, i.e., F -norm for matrices. For manifold $\mathcal{M}$, we denote Log_P , Exp_P , and $\langle \cdot, \cdot \rangle_P = g_P(\cdot, \cdot)$ as the Riemannian logarithm, exponentiation, and metric at $P \in \mathcal{M}$, respectively. Besides, $\gamma_{(P,Q)}(t)$ is the geodesic connecting P and Q with geodesic distance as $d(\cdot, \cdot)$. We provide a complete table of notations in App. A.

2.2 SPD Lie Groups

The SPD manifold has shown great success in diverse applications [9], [14], [33], [47], [48], [49]. We denote $n \times n$ SPD matrices as $\mathcal{S}_{++}^n$ and $n \times n$ real symmetric matrices as $\mathcal{S}^n$. As shown by Arsigny *et al* [36], $\mathcal{S}_{++}^n$ forms an open submanifold of the Euclidean space $\mathcal{S}^n$, known as the SPD manifold. SPD manifolds exhibit three Lie group structures, each associated with an invariant metric. These metrics include the Log-Euclidean Metric (LEM) [36], Affine-Invariant Metric (AIM) [50], and Log-Cholesky Metric (LCM) [37]. We denote LEM and AIM as (α, β) -LEM and (α, β) -AIM, as they are induced by the following $O(n)$ -invariant inner product on the tangent space at the identity matrix [51]:

$$\langle V, W \rangle^{(\alpha, \beta)} = \alpha \langle V, W \rangle + \beta \operatorname{tr}(V) \operatorname{tr}(W), \quad (3)$$

where $V, W \in T_I \mathcal{S}_{++}^n \cong \mathcal{S}^n$, and $(\alpha, \beta) \in \mathbf{ST} = \{(\alpha, \beta) \in \mathbb{R}^2 \mid \min(\alpha, \alpha + n\beta) > 0\}$. Tab. 1 summarizes the Lie structures on SPD manifolds with the following notations. Let $P, Q \in \mathcal{S}_{++}^n$ be SPD matrices. We denote the matrix logarithm, exponentiation, and Cholesky decomposition by $\log(\cdot)$, $\exp(\cdot)$, and $\text{Chol}(\cdot)$, respectively. The Cholesky factors of P and Q are $L = \text{Chol}(P)$ and $K = \text{Chol}(Q)$. $[\cdot]$ returns the strictly lower triangular part of a square matrix. Other Riemannian operators are summarized in App. B.3.

For (α, β) -LEM and LCM, the Fréchet mean admits a closed-form expression. Moreover, since (α, β) -AIM has non-positive sectional curvature [51, Tab. 5], the WFM is

2. Invariant metric always exists for every Lie group [40, Ch. 1.2].

TABLE 1: SPD Lie groups and invariant metrics.

Metric	$Q \odot P$	Invariance
(α, β) -AIM	$KPK^\top$	Left-invariance
(α, β) -LEM	$\exp(\log(P) + \log(Q))$	Bi-invariance
LCM	$\text{Chol}^{-1}([L + K] + \mathbb{K}L)$	Bi-invariance

a global operation [52, Ch. 6.1.5] and can be computed by the Karcher flow algorithm [53].

2.3 Rotation Lie Groups

The set of $n \times n$ rotation matrices form a Lie group, known as special orthogonal groups, denoted as $\text{SO}(n)$ [42]. Its group operation is the matrix product, with the identity matrix as the neutral element. Any tangent vector A in $T_R\text{SO}(n)$ can be represented as $A = RV$, with $V \in \mathfrak{so}(n)$. Here, $\mathfrak{so}(n)$ is the Lie algebra of $\text{SO}(n)$, which is the tangent space at the identity matrix, formed by the set of $n \times n$ skew-symmetric matrices. The Fréchet mean can be obtained by Karcher flow [54]. Furthermore, if all the rotations lie in a closed ball of radius $r < \pi/2$, then Karcher flow converges to the unique mean [54, Thm. 5.1]. The associated Riemannian operators are summarized in App. B.4.

2.4 Full-Rank Correlation Lie Groups

The correlation matrix of a covariance matrix Σ is defined as $C = \text{Cor}(\Sigma) = \mathbb{D}(\Sigma)^{-1/2} \Sigma \mathbb{D}(\Sigma)^{-1/2}$, where $\mathbb{D}(\cdot)$ returns a diagonal matrix with diagonal elements of Σ . The space of $n \times n$ full-rank correlation matrices, denoted as $\mathcal{C}_{++}^n$, forms a quotient manifold of the SPD manifold $\mathcal{S}_{++}^n$ [55, Thm. 1], referred to as the correlation manifold. This manifold can be interpreted as a compactly normalized SPD manifold that encodes scale-invariant information [3]. However, its Riemannian structure has been less studied than SPD matrices. Recently, Thanwerdas and Pennec [3] developed two convenient Riemannian metrics: the Euclidean-Cholesky Metric (ECM) and Log-Euclidean-Cholesky Metric (LECM). Thanwerdas [39] further proposed two permutation-invariant metrics, the Off-Log Metric (OLM) and Log-Scaled Metric (LSM). All four geometries above are pullback metrics from simpler Euclidean spaces. We first review the related prototype spaces, followed by an examination of the four Riemannian metrics.

Prototype spaces: LT_1^n and LT_0^n denote the Euclidean spaces of $n \times n$ lower triangular matrices with unit and null diagonals, respectively. Hol^n denotes the Euclidean space of $n \times n$ symmetric matrices with null diagonals. The tangent space $T_C \mathcal{C}_{++}^n$ at $C \in \mathcal{C}_{++}^n$ can be identified with Hol^n . Row_0^n denotes the Euclidean space of $n \times n$ symmetric matrices with null row sum.

ECM is derived from LT_1^n by $\mathcal{C}_{++}^n \xrightarrow[\Theta^{-1} = \text{Cor} \circ \text{Chol}^{-1}]{\Theta} \text{LT}_1^n$, where $\Theta(C) = \mathbb{D}(\text{Chol}(C))^{-1} \text{Chol}(C)$ for any $C \in \mathcal{C}_{++}^n$.

LECM is defined by further pulling back ECM: $\mathcal{C}_{++}^n \xrightarrow[\log \circ \Theta]{\log \circ \Theta} \text{LT}_0^n$. Due to the nilpotency of LT_0^n , the matrix logarithm $\log(\cdot) : \text{LT}_1^n \rightarrow \text{LT}_0^n$ and its inverse $\exp(\cdot)$ over LT_0^n are free from eigendecomposition.

OLM is derived from a permutation invariant inner product over Hol^n by $\mathcal{C}_{++}^n \xrightarrow[\text{Exp}^\circ]{\text{Log}^\circ = \text{Off} \circ \log} \text{Hol}^n$. Here, $\text{Off}(\cdot)$ returns a matrix in Hol^n consisting of off-diagonal elements. For any symmetric hollow matrix $H \in \text{Hol}^n$, there exists a unique diagonal matrix $\mathcal{D}^+(H)$, such that $\text{Exp}^\circ(H) = \exp(\mathcal{D}^+(H) + H) \in \mathcal{C}_{++}^n$ is a diffeomorphism, where $\mathcal{D}^+(H)$ can be computed by the following exponentially converged algorithm: $D_{k+1} = D_k - \log(\mathbb{D}(\exp(D_k + H)))$ [56, Sec. 5].

LSM is derived from a permutation invariant inner product over Row_0^n by $\mathcal{C}_{++}^n \xrightarrow[\text{Exp}^* = \text{Cor} \circ \exp]{\text{Log}^*} \text{Row}_0^n$. For any correlation matrix $C \in \mathcal{C}_{++}^n$, there exists a unique positive diagonal matrix $\mathcal{D}^*(C)$ such that $\text{Log}^*(C) = \log(\mathcal{D}^*(C) C \mathcal{D}^*(C)) \in \text{Row}_0^n$ is a diffeomorphism, where $\mathcal{D}^*(C)$ could be solved by damped Newton's method [39, Sec. 3.5].

Four group operations are defined as Eq. (1):

$$P \odot Q = f^{-1}(f(P) + f(Q)), \forall P, Q \in \mathcal{C}_{++}^n. \quad (4)$$

with f as Θ , $\log \circ \Theta$, Log° , and Log^* for ECM, LECM, OLM, and LSM, respectively. The following discusses the invariance and WFM. More details on Riemannian structures are presented in App. B.5.

Proposition 2.5 (Invariance & WFM). *ECM, LECM, OLM, and LSM are bi-invariant. Following the notations in Thm. 2.4, the WFM for a batch of samples $\{P_{i \dots N} \in \mathcal{C}_{++}^n\}$ are $f^{-1}(\sum_{i=1}^N w_i f(P_i))$, with f as Θ , $\log \circ \Theta$, Log° , and Log^* for ECM, LECM, OLM, and LSM, respectively.*

Proof. The proof is presented in App. H.1. $\square$

3 REVISITING NORMALIZATION

3.1 Revisiting Euclidean Normalization

In Euclidean DNNs, normalization is a significant technique for accelerating network training by mitigating the issue of internal covariate shift [28]. While various normalization methods have been introduced [28], [29], [30], [31], they all share a common purpose: the normalization of the first and second moments. This paper focuses on Batch Normalization (BN), the prototype of other normalization variants.

Given a batch of activations $\{x_{i \dots N}\}$, the core operations in the standard Euclidean BN can be expressed as:

$$\forall i \leq N, x_i \leftarrow \gamma \frac{x_i - \mu_b}{\sqrt{v_b^2 + \epsilon}} + \beta \quad (5)$$

where μ_b is the batch mean, v_b^2 is the batch variance, γ is the scaling parameter, β is the biasing parameter, and ϵ is a small scalar for stability.

3.2 Revisiting Riemannian Batch Normalization

Although endeavors have been made to develop Riemannian normalization approaches tailored for manifolds, none of the existing methods effectively handle the first and second moments in a principled manner.

TABLE 2: Summary of some representative RBN methods.

Methods	Involved Statistics	Controllable Mean	Controllable Variance	Geometries
SPDBN [32, Alg. 1]	Mean	✓	N/A	SPD manifolds under AIM
SPDBN [57, Alg. 1]	Mean+Variance	✓	✓	SPD manifolds under AIM
SPDDSMBN [33]	Mean+Variance	✓	✓	SPD manifolds under AIM
ManifoldNorm [34, Algs. 1-2]	Mean+Variance	✗	✗	Riemannian homogeneous space
ManifoldNorm [34, Algs. 3-4]	Mean+Variance	✓	✓	A specific Lie group structure and distance
RBN [35, Alg. 2]	Mean+Variance	✗	✗	Geodesically complete manifolds
LieBN (Ours)	Mean+Variance	✓	✓	General Lie groups

Brooks *et al* [32] introduced RBN over SPD manifolds under AIM. The core operations are defined as follows:

$$\text{Centering from mean } M \in \mathcal{S}_{++}^n : \bar{P}_i \leftarrow M^{-\frac{1}{2}} P_i M^{-\frac{1}{2}}, \quad (6)$$

$$\text{Biasing towards parameter } B \in \mathcal{S}_{++}^n : \hat{P}_i \leftarrow B^{\frac{1}{2}} \bar{P}_i B^{\frac{1}{2}}, \quad (7)$$

where $\{P_{i \dots N}\}$ are SPD matrices, and M are their Fréchet mean under AIM. Let $\Gamma_{P \rightarrow Q}(S) = \text{Exp}_Q[\text{PT}_{P \rightarrow Q}(\text{Log}_P(S))]$, where $P, Q, S \in \mathcal{S}_{++}^n$. Under AIM, Eqs. (6) and (7) can be more generally expressed as

$$\Gamma_{I \rightarrow B}[\Gamma_{M \rightarrow I}(P_i)]. \quad (8)$$

However, Eqs. (6) and (7) only consider the Riemannian mean³ and does not consider the Riemannian variance. To remedy this limitation, Kobler *et al* [33] further extended the RBN to involve the second-order statistics. The key operation is formulated as

$$\forall i \leq N, \bar{P}_i \leftarrow \Gamma_{I \rightarrow B}[(\Gamma_{M \rightarrow I}(P_i))^{\frac{s}{v}}], \quad (9)$$

where v^2 is the Fréchet variance, $s \in \mathbb{R}$ is a scaling factor. However, this method is still limited to SPD manifolds under AIM. In parallel, Chakraborty [34, Algs. 1-2] proposed a general framework for Riemannian homogeneous spaces based on Eq. (8), which involves both first and second moments. However, Eq. (8) does not generally guarantee the control over Riemannian mean, resulting in agnostic Riemannian statistics [34, Sec. 3.1]. To mitigate this limitation, Chakraborty [34, Algs. 3-4] further proposed normalization over the matrix Lie group. However, the discussion is limited to a certain distance, limiting the applicability of their method. On the other hand, Lou *et al* [35, Alg. 2] proposed an RBN based on a variance of Eq. (8). Similarly, their approach suffers from the same problem of agnostic Riemannian statistics on general manifolds.

In summary, prevailing Riemannian normalization approaches lack a principled guarantee for controlling the first and second-order statistics. In contrast, our method can normalize first and second-order statistics over general Lie groups. We summarize the above RBN methods in Tab. 2.

4 LIE GROUP BATCH NORMALIZATION

Since every Lie group naturally admits invariant metrics, we propose batch normalization over Lie groups based on invariant metrics, referred to as LieBN. We first introduce the core operations under left-invariant metrics and then extend them to right-invariant metrics. Finally, we present the

theoretical LieBN framework. In the following, we denote the neutral element in the Lie group $\mathcal{M}$ as E^4 .

4.1 Ingredients under Left-invariant Metrics

In this subsection, we always assume that the Lie group $\mathcal{M}$ admits a left-invariant metric g^L . Recalling the standard Euclidean BN [28] in Eq. (5), two key points are noteworthy: (a) the Euclidean BN implicitly assumes a Gaussian distribution and can effectively normalize the latent Gaussian distribution; (b) the centering and biasing operations control the mean, while the scaling controls the variance. Therefore, extending BN into Lie groups requires the counterparts of Gaussian distribution, centering, biasing, and scaling.

There are several notions of Gaussian distribution over manifolds [58], [59], [60], [61]. We adopt the intrinsic definition from Chakraborty *et al* [60], which characterizes a Gaussian distribution on the Lie group $\mathcal{M}$ with a mean parameter $M \in \mathcal{M}$ and variance σ^2 . This distribution is denoted as $\mathcal{N}(M, \sigma^2)$, and its Probability Density Function (P.D.F.) is

$$p(X | M, \sigma^2) = k(\sigma) \exp\left(-\frac{d(X, M)^2}{2\sigma^2}\right), \quad (10)$$

where $k(\sigma)$ is the normalizing constant and $d(\cdot, \cdot)$ is the geodesic distance. When $\mathcal{M}$ is $\mathbb{R}$ with the standard Euclidean metric, Eq. (10) reduces to the Euclidean Gaussian.

On Lie groups, the natural counterparts of addition and subtraction in Eq. (5) are group operations. Therefore, centering and biasing on Lie groups can be defined by the left translation. Additionally, we define scaling via the tangent space. Specifically, for a batch of activations $\{P_{i \dots N} \in \mathcal{M}\}$, we define the key operations of LieBN as follows:

$$\text{Centering from mean } M \in \mathcal{M} : \bar{P}_i \leftarrow L_{M \odot}^{-1}(P_i), \quad (11)$$

$$\text{Scaling: } \hat{P}_i \leftarrow \text{Exp}_E\left[\frac{s}{\sqrt{v^2 + \epsilon}} \text{Log}_E(\bar{P}_i)\right], \quad (12)$$

$$\text{Biasing towards parameter } B \in \mathcal{M} : \tilde{P}_i \leftarrow L_B(\hat{P}_i), \quad (13)$$

where M is the Fréchet mean, v^2 is the Fréchet variance, $M \odot^{-1} \in \mathcal{M}$ is the group inverse of M , $L_{M \odot}^{-1}$ and L_B are left translations ($L_B(P_i) = B \odot P_i$), and $s \in \mathbb{R}/\{0\}$ is a scaling parameter. The following two propositions demonstrate the above operations in normalizing mean and variance: one related to population statistics and the other related to sample statistics.

3. Although not discussed in [32], Eqs. (6) and (7) as congruent actions can transfer the batch mean into desirable one under AIM.

4. The neutral element E is not necessarily the identity matrix.

Proposition 4.1 (Population). *Given a random point X over $\{\mathcal{M}, g^L\}$, and the Gaussian distribution $\mathcal{N}(M, v^2)$ defined in Eq. (10), we have the following for the population statistics:*

- 1) (MLE of M) Given $\{P_{i \dots N} \in \mathcal{M}\}$ i.i.d. sampled from $\mathcal{N}(M, v^2)$, the maximum likelihood estimator (MLE) of M is the sample Fréchet mean.
- 2) (Gaussian homogeneity) Given $X \sim \mathcal{N}(M, v^2)$ and $B \in \mathcal{M}$, $L_B(X) \sim \mathcal{N}(L_B(M), v^2)$

Proof. The proof is presented in App. H.2. $\square$

Proposition 4.2 (Sample). *Given N samples $\{P_{i \dots N}\}$ over the Lie group $\{\mathcal{M}, g^L\}$, denoting $\phi_s(P_i) = \text{Exp}_E[s \text{Log}_E(P_i)]$, we have the following for the sample statistics.*

- Sample mean homogeneity:

$$\text{FM}\{L_B(P_i)\} = L_B(\text{FM}\{P_i\}), \forall B \in \mathcal{M}. \quad (14)$$

- Controllable dispersion from E :

$$\sum_{i=1}^N w_i d^2(\phi_s(P_i), E) = s^2 \sum_{i=1}^N w_i d^2(P_i, E), \quad (15)$$

where $\{w_{1 \dots N}\}$ are weights satisfying a convexity constraint, i.e., $\forall i, w_i > 0$ and $\sum_i w_i = 1$.

Proof. The proof is presented in App. H.3. $\square$

Thm. 4.1 and Eq. (14) imply that our centering and biasing in Eqs. (11) and (13) can transfer the sample and population mean. As the post-centering mean is E , Eq. (15) implies that Eq. (12) can control the sample variance. More interestingly, the latent Gaussian distribution can be transferred under some geometries, such as SPD manifolds under LEM and LCM, which are discussed in App. D.

Remark 4.3. The MLE of the mean of the Gaussian distribution has been examined in several previous works [34], [60], [62]. However, these studies primarily focus on particular manifolds or specific metrics. In contrast, our contribution lies in presenting a general result for Lie groups.

Remark 4.4. While Eq. (10) appeared in [57], the authors only focus on SPD manifolds under AIM. The transformation of the population under their proposed RBN remains unexplored as well. Besides, while Chakraborty [34] analyzed the population properties for their RBN over matrix Lie groups, their results were confined within a specific distance. In contrast, our work provides a more extensive examination, encompassing both population and sample properties of our LieBN in a general manner.

4.2 Ingredients under Right-invariant Metrics

The key insight beneath Eqs. (11) and (13) and Thms. 4.1 and 4.2 is that left translation is an isometry under left-invariant metrics. Similarly, right translation is an isometry under right-invariant metrics. Therefore, it can be used for centering and biasing under right-invariant metrics. Fol-

lowing the previous notations, we define the centering and biasing under a right-invariant metric g^R as

$$\text{centering to } E: \bar{P}_i \leftarrow R_{M_\odot^{-1}}(P_i), \quad (16)$$

$$\text{biasing towards } B: \tilde{P}_i \leftarrow R_B(\hat{P}_i). \quad (17)$$

Similar to the case under left-invariant metrics, Thms. 4.1 and 4.2 can be easily extended into the ones under right-invariant metrics. Notable, the proofs for MLE of M in Thm. 4.1 and controllable dispersion in Thm. 4.2 can be directly applied to the right-invariant metric. Therefore, we only show the homogeneity in the following proposition.

Proposition 4.5. *Given a random point $X \sim \mathcal{N}(M, v^2)$ over $\{\mathcal{M}, g^R\}$, $B \in \mathcal{M}$, and N samples $\{P_{i \dots N}\}$ over $\mathcal{M}$, we have*

- 1) Gaussian homogeneity: $R_B(X) \sim \mathcal{N}(R_B(M), v^2)$;
- 2) Sample homogeneity: $\text{FM}\{R_B(P_i)\} = R_B(\text{FM}\{P_i\})$.

Proof. The proof is presented in App. H.4. $\square$

4.3 LieBN under Invariant Metrics

Algorithm 1: Lie Group Batch Normalization (LieBN)

Input : A batch of activations $\{P_{1 \dots N}\}$ over Lie groups $\{\mathcal{M}, \odot, g\}$, a small positive constant ϵ , and momentum $\gamma \in [0, 1]$, running mean $M_r = E$, running variance $v_r^2 = 1$, biasing parameter $B \in \mathcal{M}$, and scaling parameter $s \in \mathbb{R}/\{0\}$.

Output : Normalized activations $\{\tilde{P}_{1 \dots N}\}$.

if training then

 Compute batch mean M_b and variance v_b^2

 Update running statistics:

$$M_r \leftarrow \text{WFM}(\{1 - \gamma, \gamma\}, \{M_r, M_b\})$$

$$v_r^2 \leftarrow (1 - \gamma)v_r^2 + \gamma v_b^2$$

 Use the batch statistics, $M \leftarrow M_b, v^2 \leftarrow v_b^2$

else

 Use the running statistics, $M \leftarrow M_r, v^2 \leftarrow v_r^2$

end

for $i \leftarrow 1$ to N do

 Centering to the neutral element E :

if g is left-invariant then

$$\quad \bar{P}_i \leftarrow L_{M_\odot^{-1}}(P_i)$$

else

$$\quad \bar{P}_i \leftarrow R_{M_\odot^{-1}}(P_i)$$

end

 Scaling the variance:

$$\hat{P}_i \leftarrow \text{Exp}_E \left[\frac{s}{\sqrt{v^2 + \epsilon}} \text{Log}_E(\bar{P}_i) \right]$$

 Biasing towards parameter B :

if g is left-invariant then

$$\quad \tilde{P}_i \leftarrow L_B(\hat{P}_i)$$

else

$$\quad \tilde{P}_i \leftarrow R_B(\hat{P}_i)$$

end

end

With the above ingredients, Alg. 1 presents our theoretical LieBN framework. Similar to Ioffe *et al* [28], we use the

moving average to update the running statistics. For a bi-invariant metric, LieBN can be implemented using either left or right translation. If the Lie group is commutative, LieBN under left and right translations are equivalent. Tab. 3 summarizes the LieBN types under different conditions.

TABLE 3: Summary of LieBN types.

Commutativity	Non-commutative			Commutative
Invariance	Left	Right	Bi	Left = Right = Bi
LieBN Types	Left	Right	Left & Right	Left = Right

The centering and biasing in Euclidean BN correspond to the group action of $\mathbb{R}$. From a geometric perspective, the standard Euclidean metric is invariant under this group operation. Consequently, it is not surprising that our LieBN algorithm naturally generalizes the standard Euclidean BN.

Proposition 4.6. *The LieBN algorithm presented in Alg. 1 is equivalent to the standard Euclidean BN when $\mathcal{M} = \mathbb{R}^n$, both during the training and testing phases.*

Proof. The proof is presented in App. H.5. $\square$

5 MANIFESTATIONS

This section instantiates our LieBN in Alg. 1 on nine different Lie groups, including four on the SPD manifold, one on the rotation, and four on the correlation manifold.

5.1 LieBN on SPD Manifolds

We first extend the current Lie groups on SPD manifolds by the matrix power deformation, resulting in three families of parameterized Lie groups. Then, we propose a novel right-invariant metric on the SPD manifold, the first non-trivial right-invariant metric on this manifold. Finally, we construct LieBN layers based on these Lie structures.

5.1.1 Deformed Lie Structures on SPD Manifolds

As shown in Tab. 1, there are three Lie groups on SPD manifolds, each with a left-invariant metric. These metrics include (α, β) -AIM, (α, β) -LEM, and LCM. For clarity, we denote the group operations w.r.t. (α, β) -AIM, (α, β) -LEM and LCM as $\odot^{\text{AI}}$, $\odot^{\text{LE}}$ and $\odot^{\text{LC}}$, respectively.

Recently, Thanwerdas and Pennec [63] further extended (α, β) -AIM into three-parameter families of metrics by the pullback of matrix power function $P_\theta(\cdot)$ and scaled by $\frac{1}{\theta^2}$, denoted as (θ, α, β) -AIM. The matrix power serves as a deformation, wherein (θ, α, β) -AIM encompasses (α, β) -AIM with $\theta = 1$, and becomes (α, β) -LEM as θ approaches 0 [64]. Inspired by the deforming utility of the power function, we define the power-deformed metrics of (α, β) -LEM and LCM as the pullback metrics by P_θ and scaled by $\frac{1}{\theta^2}$. We denote these two metrics as (θ, α, β) -LEM and θ -LCM, respectively. We have the following results w.r.t. the deformation.

Proposition 5.1 (Deformation). *(θ, α, β) -LEM is equal to (α, β) -LEM. θ -LCM interpolates between $\tilde{g}$ -LEM ($\theta = 0$) and LCM ($\theta = 1$). Here, given any $P \in \mathcal{S}_{++}^n$ and tangent vectors*

$V, W \in T_P \mathcal{S}_{++}^n$, $\tilde{g}$ -LEM is defined as

$$\langle V, W \rangle_P = \tilde{g}(\log_{*,P}(V), \log_{*,P}(W)), \quad (18)$$

where $\tilde{g}(V_1, V_2) = \frac{1}{2}\langle V_1, V_2 \rangle - \frac{1}{4}\langle \mathbb{D}(V_1), \mathbb{D}(V_2) \rangle$, $\mathbb{D}(V_i)$ is a diagonal matrix consisting of the diagonal elements of V_i , and $\log_{*,P}$ is the differential map at P .

Proof. The proof is presented in App. H.6. $\square$

As (θ, α, β) -LEM is equal to (α, β) -LEM, we focus on (α, β) -LEM, (θ, α, β) -AIM, and θ -LCM in the following. As a diffeomorphism, P_θ also can pull back the group operation $\odot^{\text{AI}}$ and $\odot^{\text{LC}}$, denoted as $\odot^{\theta\text{-AI}}$ and $\odot^{\theta\text{-LC}}$. We have the following proposition on the invariance.

Proposition 5.2 (Invariance). *(θ, α, β) -AIM is left-invariant w.r.t. $\odot^{\theta\text{-AI}}$, while θ -LCM is bi-invariant w.r.t. $\odot^{\theta\text{-LC}}$.*

Proof. The proof is presented in App. H.7. $\square$

5.1.2 SPD Right-invariant Metrics

AIM is left-invariant w.r.t. $\odot^{\text{AI}}$. We also can define right-invariant metric w.r.t. $\odot^{\text{AI}}$ by definition [40, Ch. 1.2]:

$$g_P^{\text{CRI}}(V, W) = \left\langle R_{P_{\odot^{\text{AI}}*}, P}^{-1}(V), R_{P_{\odot^{\text{AI}}*}, P}^{-1}(W) \right\rangle_I \quad (19)$$

where $R_{(\cdot)}$ denotes Lie group right translation, $P_{\odot^{\text{AI}}}^{-1}$ is the inverse of P under $\odot^{\text{AI}}$, and $\langle \cdot, \cdot \rangle_I$ denotes an arbitrary inner product on $T_I \mathcal{S}_{++}^n$. We set $\langle \cdot, \cdot \rangle_I$ the same as the AIM at I , i.e., $\langle \cdot, \cdot \rangle^{(\alpha, \beta)}$. We call this metric the Cholesky Right Invariant Metric (CRIM), as the group operation is defined by the matrix product of Cholesky factors [3, Sec. 3.2].

Theorem 5.3. *Given any SPD matrices P, Q and tangent vector $V \in T_P \mathcal{S}_{++}^n$, the Riemannian operators on $\{\mathcal{S}_{++}^n, g^{\text{CRI}}\}$ are*

$$g_P^{\text{CRI}}(V, V) = \left(\left\| \left(L(L^{-1}VL^{-\top})_{\frac{1}{2}}L^{-1} \right)_{\text{Sym}} \right\|^{(\alpha, \beta)} \right)^2 \quad (20)$$

$$d(P, Q) = \left\| \log \left(\tilde{Q}^{-\frac{1}{2}} \tilde{P} \tilde{Q}^{-\frac{1}{2}} \right) \right\|^{(\alpha, \beta)}, \quad (21)$$

$$\text{Exp}_P(V) = \left(\text{Exp}_P^{\text{AI}}(-\bar{V}) \right)_{\odot^{\text{AI}}}^{-1}, \quad (22)$$

$$\text{Log}_P(Q) = - \left(LL^\top \left(L\tilde{V}L^\top \right)_{\frac{1}{2}} \right)_{\text{Sym}}, \quad (23)$$

where L is the Cholesky factor of $P = LL^\top$, $(\cdot)_{\odot^{\text{AI}}}^{-1}$ is the group inverse, $\tilde{Q}$ and $\tilde{P}$ are the group inverses of P and Q , $\bar{V} = \left((L^{-1}VL^{-\top})_{\frac{1}{2}}L^{-1}L^{-\top} \right)_{\text{Sym}}$, and $\tilde{V} = \text{Log}_P^{\text{AI}}(\tilde{Q})$. Here, $(X)_{\text{Sym}} = X + X^\top$, $\forall X \in \mathbb{R}^{n \times n}$ denotes symmetrization, and $(\cdot)_{\frac{1}{2}}$ as its inverse map, namely $(X)_{\frac{1}{2}} = [X] + \frac{1}{2}\mathbb{X}$.

Proof. The proof is presented in App. H.8. $\square$

Corollary 5.4. *CRIM is geodesically complete, and the associated*

TABLE 4: Key operators in calculating LieBN on SPD manifolds. The notations follow App. B.3 and Thm. 5.3.

Metric		(θ, α, β) -AIM	(α, β) -LEM	θ -LCM	θ -CRIM
Invariance		Left-invariance	Bi-invariance		Right-invariance
LieBN Type		LieBN-Left	LieBN-Left = LieBN-Right		LieBN-Right
Pullback Map		P_θ	log	$P_\theta \circ \psi_{LC}$	P_θ
Codomain		$\{S_{++}^n, \odot^{AI}, \frac{1}{\theta^2} g^{(\alpha, \beta)-AI}\}$	$\{S^n, \langle \cdot, \cdot \rangle^{(\alpha, \beta)}\}$	$\{LT^n, \frac{1}{\theta^2} \langle \cdot, \cdot \rangle\}$	$\{S_{++}^n, \odot^{AI}, \frac{1}{\theta^2} g^{CRI}\}$
Riemannian and Lie group operators in the codomain	$L_Q(P)$ or $R_Q(P)$	$KPK^\top$	$P + Q$	$P + Q$	$LQL^\top$
	$L_{Q_\odot^{-1}}(P)$ or $R_{Q_\odot^{-1}}(P)$	$K^{-1}PK^{-\top}$	$P - Q$	$P - Q$	$L^{-1}QL^{-\top}$
	$\text{Exp}_E[s \text{Log}_E(P)]$	P^s	sP	sP	$\left(\left(P_{\odot^{AI}}^{-1}\right)^s\right)^{-1}_{\odot^{AI}}$
	FM	Karcher Flow	Arithmetic average	Arithmetic average	Karcher Flow
	WFM($\{1 - \gamma, \gamma\}, \{P_1, P_2\}$)	$P_1^{\frac{1}{2}} \left(P_1^{-\frac{1}{2}} P_2 P_1^{-\frac{1}{2}}\right)^\gamma P_1^{\frac{1}{2}}$	Arithmetic weighted average	Arithmetic weighted average	$\left(\tilde{P}_1^{\frac{1}{2}} \left(\tilde{P}_1^{-\frac{1}{2}} \tilde{P}_2 \tilde{P}_1^{-\frac{1}{2}}\right)^\gamma \tilde{P}_1^{\frac{1}{2}}\right)^{-1}_{\odot^{AI}}$

TABLE 5: Key operators in calculating LieBN on the rotation matrices. The notations follow App. B.4.

Invariance	LieBN Type	$R_{\odot}^{-1}$	$L_R(S)$	$R_R(S)$	$\text{Exp}_I[s \text{Log}_I(R)]$	FM	WFM($\{1 - \gamma, \gamma\}, \{R, S\}$)
Bi-invariance	LieBN-Left & LieBN-Right	R^{-1}	RS	SR	$\exp(t \log(R))$	[54, Alg. 1]	$R \exp(\gamma \log(R^\top S))$

geodesic connecting SPD matrices P and Q is

$$\begin{aligned} \gamma_{(P,Q)}(t) &= \left\{ \gamma^{AI}(t; \tilde{P}, \tilde{Q}) \right\}_{\odot^{AI}}^{-1} \\ &= \left\{ \tilde{P}^{\frac{1}{2}} \left(\tilde{P}^{-\frac{1}{2}} \tilde{Q} \tilde{P}^{-\frac{1}{2}} \right)^t \tilde{P}^{\frac{1}{2}} \right\}_{\odot^{AI}}^{-1}, \end{aligned} \quad (24)$$

where $\tilde{P} = P_{\odot^{AI}}^{-1}$ and $\tilde{Q} = Q_{\odot^{AI}}^{-1}$ are group inverses, with γ^{AI} as the geodesic under AIM.

Proof. The proof is presented in App. H.9. $\square$

Similar to the discussion in Sec. 5.1.1, we define θ -CRIM as the deformed metric of CRIM by the pullback of matrix power function $P_\theta(\cdot)$ and scaled by $\frac{1}{\theta^2}$. As the pullback of CRIM, θ -CRIM is right-invariant w.r.t. $\odot^{\theta-AI}$ by definition.

Proposition 5.5. θ -CRIM is right-invariant w.r.t. $\odot^{\theta-AI}$.

5.1.3 Manifestations on SPD Manifolds

As discussed in Secs. 5.1.1 and 5.1.2, there are four families of invariant metrics on the SPD Lie groups: (1) left-invariant (θ, α, β) -AIM w.r.t. $\odot^{AI}$; (2) bi-invariant (α, β) -LEM w.r.t. $\odot^{LE}$ and θ -LCM w.r.t. $\odot^{LC}$; (3) right-invariant θ -CRIM w.r.t. $\odot^{AI}$. Since all the above metrics are pullback metrics, the LieBN based on these metrics can be simplified and calculated in the co-domain. We first show a general result on LieBN under the pullback metric. We denote Alg. 1 on the Lie group $\mathcal{M}$ as

$$\text{LieBN}(P_i; B, s, \epsilon, \gamma), \forall P_i \in \{P_1 \dots P_N \in \mathcal{M}\}. \quad (25)$$

Then we can obtain the following theorem.

Theorem 5.6. Given a Lie group $\mathcal{M}_1$, a Lie group $\mathcal{M}_2$ with an invariant metric g^2 , and a diffeomorphism $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, then f induces an invariant metric g^1 on $\mathcal{M}_1$, denoted as $g^1 = f^* g^2$. For a batch of activation $\{P_1 \dots P_N\}$ in $\mathcal{M}_1$, $\text{LieBN}^1(P_i; B, s, \epsilon, \gamma)$

in $\mathcal{M}_1$ can be calculated in $\mathcal{M}_2$ by the following process:

Mapping data into $\mathcal{M}_2$: $\bar{P}_i = f(P_i), \bar{B} = f(B)$, (26)

Performing LieBN in $\mathcal{M}_2$: $\hat{P}_i = \text{LieBN}^2(\bar{P}_i; \bar{B}, s, \epsilon, \gamma)$, (27)

Mapping the resulting data back to $\mathcal{M}_1$: $\tilde{P}_i = f^{-1}(\hat{P}_i)$, (28)

where LieBN^2 is the LieBN on $\mathcal{M}^2$.

Proof. The proof is presented in App. H.10. $\square$

Given a metric g on S_{++}^n , the power-deformed metric $\tilde{g} = \frac{1}{\theta^2} P_\theta^* g$ is equal to $P_\theta^* (\frac{1}{\theta^2} g)$. Thm. 5.6 indicates that the LieBN under $\tilde{g}$ can be calculated by the LieBN under $\frac{1}{\theta^2} g$. Besides, as the Christoffel symbols remain the same under constant scaling, the LieBNs under $\frac{1}{\theta^2} g$ and g only differ in the variance. We denote $g^{(\alpha, \beta)-AI}$ and $g^{(\theta, \alpha, \beta)-AI}$ as the metric tensors of (α, β) -AIM and (θ, α, β) -AIM, respectively. Based on the above discussions, the computations of the LieBN under $g^{(\theta, \alpha, \beta)-AI}$ are reduced to the LieBN under $\frac{1}{\theta^2} g^{(\alpha, \beta)-AI}$. Similarly, denoting g^{CRI} and $g^{\theta-CRI}$ as the metric tensors of CRIM and θ -CRIM, then the LieBN under θ -CRIM can be calculated by the one under $\frac{1}{\theta^2} g^{CRI}$. Furthermore, as shown by Chen *et al* [22], (α, β) -LEM is a pullback metric from the Euclidean space S^n of symmetric matrices, while θ -LCM is a pullback metric from the Euclidean space LT^n of lower triangular matrices. As shown in Thm. 4.6, the LieBN in the Euclidean space S^n or LT^n is simplified to the standard Euclidean BN. Therefore, the LieBNs under (α, β) -LEM and θ -LCM can be calculated by the Euclidean BN over S^n and LT^n , respectively.

We denote the LieBN under left and right translations as LieBN-Left and LieBN-Right, respectively. Then, the LieBNs under (θ, α, β) -AIM and θ -CRIM correspond to LieBN-Left and LieBN-Right, respectively. As $\odot^{LC}$ and $\odot^{LE}$ are commutative, the LieBN-Left and LieBN-Right under (α, β) -LEM, and θ -LCM are equivalent. We denote P, Q, P_1 and P_2 as points in the codomain, i.e., S_{++}^n with scaled CRIM for θ -CRIM, S_{++}^n with scaled (α, β) -AIM for (θ, α, β) -AIM, S^n for (α, β) -LEM, and LT^n for θ -LCM,

respectively. For CRIM, we denote $\tilde{P}_i = (P_i)_{\odot \text{AI}}^{-1}$ for $i = 1, 2$. We summarize all the necessary ingredients in Tab. 4 for calculating SPD LieBN. Note that for (θ, α, β) -AIM, our scaling operation defined in Eq. (12) encompasses the scaling operation in [57, Eq. (9)] as a special case, when $(\theta, \alpha, \beta) = (1, 1, 0)$.

5.2 LieBN on Rotation Matrices

As the Riemannian metric on the rotation matrices is bi-invariant, there are two instantiations of LieBN on this manifold, *i.e.*, LieBN-Left based on the left translation and LieBN-Right based on the right translation. Particularly, the scaling can be further simplified: $\text{Exp}_I(t \text{Log}_I(R)) = \exp(t \log(R))$. For the specific $\text{SO}(3)$, the matrix exp and log can be efficiently calculated without matrix decomposition [2, Sec 3.2]. Tab. 5 presents the expressions of the required operators in Alg. 1.

5.3 LieBN on Full-Rank Correlation Matrices

TABLE 6: Summary of LieBN on the correlation. $\langle \cdot, \cdot \rangle^{(\alpha, \beta, \gamma)}$ and $\langle \cdot, \cdot \rangle^{(\alpha, \delta, \zeta)}$ are permutation-invariant inner products, which are discussed in App. B.5.2.

Metric	ECM	LECM	OLM	LSM
Invariance	Bi-invariance			
LieBN Type	LieBN-Left = LieBN-Right			
Pullback Map	Θ	$\log \circ \Theta$	Log°	Log^*
Codomain	$\{\text{LT}_1^n, \langle \cdot, \cdot \rangle\}$	$\{\text{LT}_0^n, \langle \cdot, \cdot \rangle\}$	$\{\text{Hol}^n, \langle \cdot, \cdot \rangle^{(\alpha, \beta, \gamma)}\}$	$\{\text{Row}_0^n, \langle \cdot, \cdot \rangle^{(\alpha, \delta, \zeta)}\}$

As discussed in Sec. 2.4, all four correlation metrics are bi-invariant, and their associated Lie groups are commutative. Consequently, LieBN-Left is identical to LieBN-Right. Moreover, all four correlation metrics are pullback metrics from simpler Euclidean space. Therefore, LieBN over the correlation can be implemented as Thm. 5.6: (1) map the correlation into the prototype Euclidean space, (2) apply Euclidean BN, and (3) map back to the correlation.

Optimization. Lastly, let us talk about the optimization of the correlation-valued biasing parameter $B \in \mathcal{C}_{++}^n$. As shown by [3, Sec. 4.1], the correlation matrix can be identified by the product of hyperbolic spaces via the Cholesky decomposition. Given $C \in \mathcal{C}_{++}^n$, the k -th row of the Cholesky factor $L = \text{Chol}(C)$ is $(L_{k1}, \dots, L_{k,k-1}, L_{kk}, 0, \dots, 0)$ with $L_{kk} > 0$, which belongs to the hyperbolic space of open hemisphere $\text{HS}^{k-1} = \{x \in \mathbb{R}^k \mid \|x\| = 1, x_k > 0\}$. Besides, the open hemisphere HS^n is isometric to the Poincaré ball $\mathbb{P}^n = \{x \in \mathbb{R}^n \mid \|x\| < 1\}$ by $\pi_{\text{HS}^n \rightarrow \mathbb{P}^n}((x^\top, x_{n+1})^\top) = \frac{x}{1+x_{n+1}}$. Therefore, each correlation can be parameterized with $n-1$ Poincaré vectors. Each Poincaré vector can be optimized by the well-studied Riemannian optimization [65]. The above process can be expressed as

$$C \mapsto \begin{pmatrix} 1 & 0 & \cdots & 0 \\ L_{21} & L_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & \cdots & L_{nn} \end{pmatrix} \mapsto \begin{pmatrix} x_1 \in \mathbb{P}^1 \\ \vdots \\ x_{n-1} \in \mathbb{P}^{n-1} \end{pmatrix}. \quad (29)$$

6 EXPERIMENTS

This section validates our LieBN on nine invariant metrics across the SPD, rotation, and correlation matrices.

6.1 Experiments of LieBN on the SPD Manifold

Note that our LieBN layers are architecture-agnostic and can be applied to any existing SPD neural network. Following the previous work [9], [32], [33], we focus on two network architectures: (1) SPDNet [9] for drone recognition on the Radar dataset [32], and human action recognition on the HDM05 [66] and FPHA [67] datasets; (2) TSMNet [33] for EEG classification on the Hinss2021 dataset [68]. In the EEG application, TSMNet is endowed with SPD domain-specific momentum batch normalization (TSMNet+SPDDSMNB) [33], which is a domain adaptation version of [57]. For a fair comparison, we also implement a domain-specific momentum LieBN, referred to as DSMLieBN (detailed in App. E). The backbone network architectures are represented as $\{d_0, d_1, \dots, d_L\}$, where the dimension of the parameter in the i -th BiMap layer (App. C) is $d_i \times d_{i-1}$. As (α, β) only affects variance calculation throughout LieBN, we simply set $(\alpha, \beta) = (1, 0)$ and only tune the deformation factor θ . For each family of LieBN or DSMLieBN, we report two representatives: the standard one induced from the standard metric ($\theta = 1$), and the one induced from the deformed metric with proper θ . If the standard one is already saturated, we only report the results of the standard ones. More details on implementation, datasets, and hyperparameter (θ, α, β) are presented in App. G.1.

Application to SPDNet: As SPDNet is the most classic SPD network, we apply our LieBN to SPDNet on the Radar, HDM05, and FPHA datasets. Additionally, we compare our method with SPDNetBN, which applies the SPDBN in Eqs. (6) and (7) to SPDNet. Following [32], [69], we use the architectures of $\{20, 16, 8\}$, $\{93, 30\}$, and $\{63, 33\}$ for the Radar, HDM05 and FPHA datasets, respectively. The 10-fold average results, including the average training time (s/epoch), are summarized in Tab. 7. We have three key observations regarding the choice of metrics, deformation, and training efficiency.

- **The choice of metrics:** The metric that yields the most effective LieBN layer differs for each dataset. Specifically, the optimal LieBN layers on these three datasets are the ones induced by AIM-(1), LCM-(0.5), and AIM-(1.5), respectively, **which improves the performance of SPDNet by 2.22%, 11.71%, and 4.8%**. Additionally, although the LCM-based LieBN performs worse than other LieBN variants on the Radar and FPHA datasets, it exhibits the best performance on the HDM05 dataset. These observations highlight the advantage of the generality of our LieBN framework.
- **The effect of deformation:** Deformation patterns also vary across datasets. Firstly, the standard AIM and CRIM are already saturated on the Radar dataset. Secondly, the appropriate deformation θ can further enhance the performance of LieBN. Notably, even though the LieBNs induced by LCM-(1) and CRIM-(1) impede the learning of SPDNet on the FPHA datasets, they can improve the performance under an appropriate deformation θ . These findings highlight the efficacy of the deforming geometry on the SPD manifold.
- **Efficiency:** Although our LieBN involves additional

TABLE 7: 10-fold average results of SPDNet with and without SPDBN or LieBN on the Radar, HDM05, and FPHA datasets. If the LieBN under the standard metric ($\theta = 1$) is not saturated, the rightmost columns report the deformed LieBN.

(a) Radar dataset.

Acc	SPDNet	SPDNetBN	SPDNetLieBN				
			$\theta = 1$				Best θ
			AIM-(1)	LEM-(1)	LCM-(1)	CRIM-(1)	LCM-(-0.5)
Fit time (s)	0.98	1.56	1.62	1.28	1.11	1.82	1.43
Mean $\pm$ STD	93.25 $\pm$ 1.10	94.85 $\pm$ 0.99	95.47$\pm$0.90	94.89 $\pm$ 1.04	93.52 $\pm$ 1.07	94.35 $\pm$ 0.68	94.80 $\pm$ 0.71
Max	94.4	96.13	96.27	96.8	95.2	95.6	95.73

(b) HDM05 dataset.

Acc	SPDNet	SPDNetBN	SPDNetLieBN						
			$\theta = 1$				Best θ		
			AIM-(1)	LEM-(1)	LCM-(1)	CRIM-(1)	AIM-(1.5)	LCM-(0.5)	CRIM-(0.5)
Fit time (s)	0.57	0.97	1.14	0.87	0.66	1.37	1.46	1.01	1.74
Mean $\pm$ STD	59.13 $\pm$ 0.67	66.72 $\pm$ 0.52	67.79 $\pm$ 0.65	65.05 $\pm$ 0.63	66.68 $\pm$ 0.71	63.25 $\pm$ 0.88	68.16 $\pm$ 0.68	70.84$\pm$0.92	65.76 $\pm$ 0.54
Max	60.34	67.66	68.75	66.05	68.52	64.94	69.25	72.27	66.96

(c) FPHA dataset.

Acc	SPDNet	SPDNetBN	SPDNetLieBN						
			$\theta = 1$				Best θ		
			AIM-(1)	LEM-(1)	LCM-(1)	CRIM-(1)	AIM-(1.5)	LCM-(0.5)	CRIM-(-0.5)
Fit time (s)	0.32	0.62	0.8	0.55	0.39	0.92	1.03	0.65	1.21
Mean $\pm$ STD	85.59 $\pm$ 0.72	89.33 $\pm$ 0.49	89.70 $\pm$ 0.51	86.56 $\pm$ 0.79	77.64 $\pm$ 1.00	84.65 $\pm$ 1.20	90.39$\pm$0.66	86.33 $\pm$ 0.43	86.40 $\pm$ 0.57
Max	86	90.17	90.5	87.83	79	86.67	92.17	87	87.17

computations on variance compared with SPDNetBN, our LieBN achieves comparable or even better efficiency than SPDNetBN. Particularly, the LieBN induced by standard LEM or LCM exhibits better efficiency than SPDNetBN. Even with deformation, the LCM-based LieBN is still comparable with SPDNetBN in terms of efficiency. This phenomenon could be attributed to the fast and simple computation of LCM and LEM.

TABLE 8: Cross-validation results of TSMNet with SPDDSMBN and DSMLieBN on the Hinss dataset. If the DSMLieBN under the standard metric ($\theta = 1$) is not saturated, the bottom rows report deformed DSMLieBN.

(a) Inter-session classification

Method		Fit Time	Mean $\pm$ STD
SPDDSMBN		0.16	54.12 $\pm$ 9.87
DSMLieBN	AIM-(1)	0.16	55.10$\pm$7.61
	LEM-(1)	0.13	54.95 $\pm$ 10.09
	LCM-(1)	0.10	51.54 $\pm$ 6.88
	CRIM-(1)	0.29	51.86 $\pm$ 9.21
	LCM-(0.5)	0.15	53.11 $\pm$ 5.65

(b) Inter-subject classification

Method		Fit Time	Mean $\pm$ STD
SPDDSMBN		7.74	50.10 $\pm$ 8.08
DSMLieBN	AIM-(1)	6.94	50.04 $\pm$ 8.01
	LEM-(1)	4.71	50.95 $\pm$ 6.40
	LCM-(1)	3.59	51.86 $\pm$ 4.53
	CRIM-(1)	16.35	50.71 $\pm$ 8.1
	CRIM-(1.5)	19.51	51.34 $\pm$ 5.82
	AIM-(-0.5)	8.71	53.97$\pm$8.78

Application to EEG classification: We apply our method to TSMNet under two scenarios, inter-session and inter-subject. Following [33], we adopt the architecture of $\{40, 20\}$. Compared to SPDDSMBN, DSMLieBN-AIM obtains the highest average scores of 55.10% and 53.97% in these two scenarios, **outperforming SPDDSMBN by 0.98% and 3.87%, respectively**. In the inter-subject scenario, the advantage of the efficiency of our LieBN over SPDDSMBN is more obvious. Specifically, both the LEM- and LCM-based DSMLieBN achieve similar or better performance compared to SPDDSMBN, while requiring considerably less training time. For example, DSMLieBN-LCM-(1) achieves better results with only half the training time of SPDDSMBN on inter-subject tasks. Interestingly, under the standard AIM, the sole difference between SPDDSMBN and our DSMLieBN is the way of centering and biasing. SPDDSMBN applies the matrix inverse square root and matrix square root to fulfill centering and biasing, while AIM-induced LieBN uses more efficient Cholesky decomposition. As such, the DSMLieBN induced by the standard AIM is more efficient than SPDDSMBN, particularly on the inter-subject task. On the other hand, the CRIM-based LieBN shows less efficiency, due to the relatively complex Riemannian computation of this metric.

Visualization: We randomly select 50 samples and visualize the input and output of LieBN on the HDM05 dataset. Using Riemannian t-SNE [70], we map the 30×30 SPD matrices to 2×2 low-dimensional representations. As shown in Fig. 3, LieBN effectively normalizes the data distribution. Specifically, the input t-SNE embeddings are largely scattered and their elements can reach up to 400K, whereas those of the output embeddings are mostly constrained within 20.

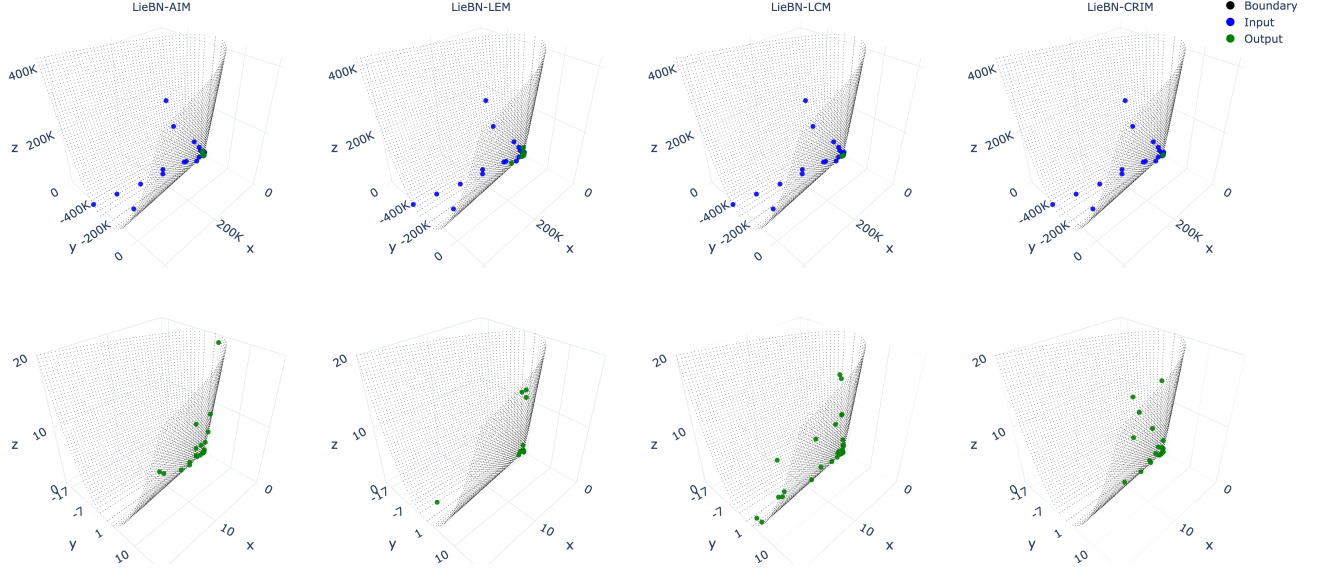


Fig. 3: Visualization of input and output 30×30 SPD matrices in LieBN using 2×2 Riemannian t-SNE embeddings. The first row shows the input and output under different metrics. Due to the significant difference in magnitude between the t-SNE embeddings of LieBN’s input and output, the second row separately visualizes the LieBN output (at a smaller scale).

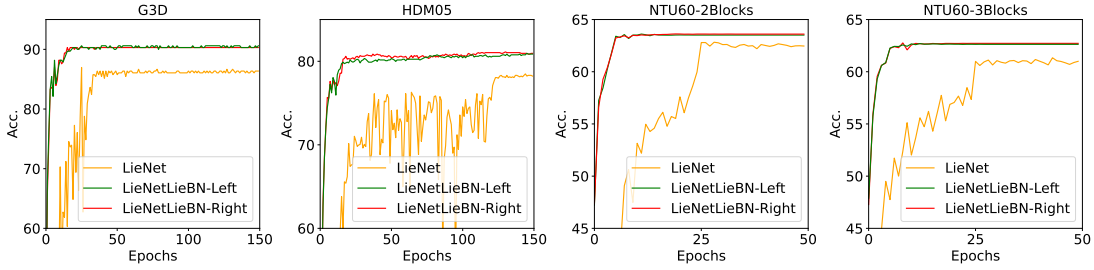


Fig. 4: Test accuracy curves corresponding to Tab. 9.

6.2 Experiments of LieBN on Rotation Matrices

This subsection implements our LieBN on the special orthogonal groups, *i.e.*, $SO(n)$, also known as rotation matrices. As the Riemannian metric on $SO(n)$ is bi-invariant, there are two instantiations of our LieBN on this group: LieBN-Left based on the left translation and LieBN-Right based on the right translation. We apply our LieBN to the classic LieNet backbone [10], where the latent space is the special orthogonal group. Following [10], we use three action recognition datasets, the G3D [71], HDM05 [66], and NTU60 [72] datasets. We denote the LieNet models with our LieBN-Left and LieBN-Right as LieNetLieBN-Left and LieNetLieBN-Right, respectively. More implementation details are presented in App. G.2.

TABLE 9: Results of LieNet with or without rotation LieBN.

Method	G3D		HDM05		NTU60	
	Mean $\pm$ STD	Max	Mean $\pm$ STD	Max	2Blocks	3Blocks
LieNet	87.91 $\pm$ 0.90	89.73	76.92 $\pm$ 1.27	79.11	62.4	60.91
LieNetLieBN-Left	88.88$\pm$1.62	90.67	78.89 $\pm$ 1.07	80.88	63.51	62.62
LieNetLieBN-Right	88.12 $\pm$ 1.12	90.3	79.39$\pm$1.13	80.67	63.6	62.72

Results: We conduct 10-fold experiments on the G3D and

HDM05 datasets under the suggested 3Blocks⁵ and 2Blocks architectures, respectively. On the NTU60 dataset, we validate LieBN under the 2Blocks and 3Blocks settings. The results are presented in Tab. 9. Due to differences in software, our reimplemented LieNet (by PyTorch) performs slightly differently from the results reported in [10] (by Matlab). However, we still observe a clear improvement when applying our LieBN to the vanilla LieNet backbone. Additionally, LieBN-Right performs slightly better than LieBN-Left. Although the effects of left and right translations on the sample statistics under the bi-invariant metric are identical, their transformations on each sample point, as illustrated in Fig. 1. This difference could slightly affect the network performance. The specific optimal choice of left or right translations depends on the dataset’s characteristics.

Training Dynamics: Fig. 4 presents the test accuracy curve. We have the following additional observations, which can be attributed to the mitigated covariate shift by our LieBN, as our LieBN can effectively normalize the sample statistics. **Accelerated convergence:** LieBN significantly accelerates the convergence of LieNet. Specifically, on the NTU60 dataset—the largest dataset involved—LieNet with LieBN

5. Each block consists of a RotMap layer followed by a RotPooling layer. For more details, please refer to [10].

converges by the 5th epoch, whereas the vanilla LieNet does not converge until the 25th epoch. A similar phenomenon can also be observed on the HDM05 dataset. **Stabler performance:** LieBN enhances the stability of network training. Especially on the HDM05 and G3D datasets, the initial training fluctuations are greatly mitigated by our LieBN.

6.3 Experiments of LieBN on Correlation Matrices

As no neural networks are specifically designed for the correlation manifold and correlation matrices are still SPD, we apply our correlation LieBN (LieBN-Cor) to SPD networks. Our experiments focus on the SPDNet backbone using the FPHA and HDM05 datasets. LieBN-Cor is applied before the final classification layer. Specifically, SPD features are first activated by the power function, then mapped into correlation matrices via $\text{Cor}(\cdot)$, and finally processed by LieBN-Cor. More details can be found in App. G.3.

TABLE 10: Results of SPDNet with or without correlation LieBN under different invariant metrics.

Dataset	SPDNet	SPDNetLieBN-Cor			
		ECM	LECM	OLM	LSM
HDM05	59.13±0.67	65.37 ± 1.07	61.35 ± 0.34	60.33 ± 0.12	60.00 ± 0.27
FPHA	85.59±0.72	87.20 ± 0.12	87.03 ± 0.32	86.80 ± 0.12	86.77 ± 0.29

Result: The 5-fold average results are presented in Tab. 10. Although LieBN-Cor is not specifically designed for SPD networks, it still improves SPDNet’s performance, demonstrating its effectiveness. Among the four invariant metrics, ECM achieves the best performance. While LieBN-SPD outperforms LieBN-Cor when applied to SPDNet—an expected result since SPDNet is tailored for SPD matrices—this does not undermine the validity of LieBN-Cor. The consistent improvement over vanilla SPDNet highlights the potential of applying LieBN-Cor to correlation manifolds.

TABLE 11: Ablations on optimizing correlation parameter in LieBN under different metrics.

Metric	ECM	LECM	OLM	LSM
Optim				
Trivialization	63.17 ± 1.32	58.84 ± 0.51	59.84 ± 0.45	53.22 ± 1.62
Riemannian	65.37 ± 1.07	61.35 ± 0.34	60.33 ± 0.12	60.00 ± 0.27

Ablations: As discussed in Sec. 5.3, the correlation-valued biasing parameter $B \in \mathcal{C}_{++}^n$ is optimized via Riemannian optimization over multiple Poincaré vectors. Alternatively, trivialization tricks [73] can be employed. Specifically, Thm. 5.6 suggests that we can instead set $\tilde{B} = f(B) \in V$ as the parameter, where f is the isometry and V is the prototype space for ECM, LECM, OLM, and LSM, respectively. Tab. 11 presents the 5-fold comparison, demonstrating the superiority of our Poincaré parameterization.

7 CONCLUSIONS

This paper presents a novel LieBN framework for batch normalization over Lie groups, leveraging natural Lie-group-invariant metrics. Compared to prior approaches, LieBN provides a principled method to normalize both sample and population statistics. Empirically, we generalize three

existing Lie group structures on the SPD manifold and introduce the first non-trivial right-invariant SPD metric. By employing these parameterized invariant metrics, we instantiate our framework on the SPD manifold. Furthermore, we implement LieBN on rotation matrices using a bi-invariant metric and on the correlation manifold using four bi-invariant metrics. Extensive experiments across different manifolds validate the effectiveness of our LieBN.

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REFERENCES

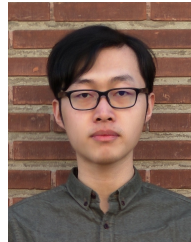
- [1] O. Yair, M. Ben-Chen, and R. Talmon, “Parallel transport on the cone manifold of SPD matrices for domain adaptation,” *IEEE TIP*, 2019.
- [2] R. Hartley, J. Trumpf, Y. Dai, and H. Li, “Rotation averaging,” *IJCV*, 2013.
- [3] Y. Thanwerdas and X. Pennec, “Theoretically and computationally convenient geometries on full-rank correlation matrices,” *SIMAX*, 2022.
- [4] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [5] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” in *NeurIPS*, 2012.
- [6] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *CVPR*, 2016.
- [7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *NeurIPS*, 2017.
- [8] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, “Geometric deep learning: going beyond Euclidean data,” *IEEE Signal Processing Magazine*, vol. 34, no. 4, pp. 18–42, 2017.
- [9] Z. Huang and L. Van Gool, “A Riemannian network for SPD matrix learning,” in *AAAI*, 2017.
- [10] Z. Huang, C. Wan, T. Probst, and L. Van Gool, “Deep learning on Lie groups for skeleton-based action recognition,” in *CVPR*, 2017.
- [11] Z. Huang, J. Wu, and L. Van Gool, “Building deep networks on Grassmann manifolds,” in *AAAI*, 2018.
- [12] O. Ganea, G. Bécigneul, and T. Hofmann, “Hyperbolic neural networks,” in *NeurIPS*, 2018.
- [13] R. Chakraborty, C.-H. Yang, X. Zhen, M. Banerjee, D. Archer, D. Vaillancourt, V. Singh, and B. Vemuri, “A statistical recurrent model on the manifold of symmetric positive definite matrices,” in *NeurIPS*, 2018.
- [14] R. Chakraborty, J. Bouza, J. H. Manton, and B. C. Vemuri, “Manifoldnet: A deep neural network for manifold-valued data with applications,” *IEEE TPAMI*, 2020.
- [15] R. Wang, X.-J. Wu, Z. Chen, T. Xu, and J. Kittler, “Learning a discriminative SPD manifold neural network for image set classification,” *Neural Networks*, vol. 151, pp. 94–110, 2022.
- [16] V. De Bortoli, E. Mathieu, M. Hutchinson, J. Thornton, Y. W. Teh, and A. Doucet, “Riemannian score-based generative modelling,” in *NeurIPS*, 2022.

- [17] R. Wang, X.-J. Wu, Z. Chen, T. Xu, and J. Kittler, "DreamNet: A deep Riemannian manifold network for SPD matrix learning," in *ACCV*, 2022.
- [18] Z. Chen, T. Xu, X.-J. Wu, R. Wang, Z. Huang, and J. Kittler, "Riemannian local mechanism for SPD neural networks," in *AAAI*, 2023.
- [19] K. Chen, J. Song, S. Liu, N. Yu, Z. Feng, G. Han, and M. Song, "Distribution knowledge embedding for graph pooling," *IEEE TKDE*, 2023.
- [20] J. Yim, B. L. Trippe, V. De Bortoli, E. Mathieu, A. Doucet, R. Barzilay, and T. Jaakkola, "SE (3) diffusion model with application to protein backbone generation," in *ICML*, 2023.
- [21] Z. Chen, Y. Song, X.-J. Wu, and N. Sebe, "Product geometries on Cholesky manifolds with applications to SPD manifolds," *arXiv preprint arXiv:2407.02607*, 2024.
- [22] Z. Chen, Y. Song, G. Liu, R. R. Kompella, X. Wu, and N. Sebe, "Riemannian multiclass logistics regression for SPD neural networks," in *CVPR*, 2024.
- [23] R. Wang, X.-J. Wu, Z. Chen, C. Hu, and J. Kittler, "SPD manifold deep metric learning for image set classification," *IEEE TNNLS*, 2024.
- [24] R. Wang, C. Hu, Z. Chen, X.-J. Wu, and X. Song, "A Grassmannian manifold self-attention network for signal classification," in *IJCAI*, 2024.
- [25] X. S. Nguyen, S. Yang, and A. Histace, "Matrix manifold neural networks++," in *ICLR*, 2024.
- [26] Z. Chen, Y. Song, Xiaojun, and N. Sebe, "RMLR: Extending multinomial logistic regression into general geometries," in *NeurIPS*, 2024.
- [27] Z. Chen, Y. Song, X. Wu, G. Liu, and N. Sebe, "Understanding matrix function normalizations in covariance pooling through the lens of Riemannian geometry," in *ICLR*, 2025.
- [28] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *ICML*, 2015.
- [29] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
- [30] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: the missing ingredient for fast stylization," *arXiv preprint arXiv:1607.08022*, 2016.
- [31] Y. Wu and K. He, "Group normalization," in *ECCV*, 2018.
- [32] D. Brooks, O. Schwander, F. Barbaresco, J.-Y. Schneider, and M. Cord, "Riemannian batch normalization for SPD neural networks," in *NeurIPS*, 2019.
- [33] R. Kobler, J.-i. Hirayama, Q. Zhao, and M. Kawanabe, "SPD domain-specific batch normalization to crack interpretable unsupervised domain adaptation in EEG," in *NeurIPS*, 2022.
- [34] R. Chakraborty, "ManifoldNorm: Extending normalizations on Riemannian manifolds," *arXiv preprint arXiv:2003.13869*, 2020.
- [35] A. Lou, I. Katsman, Q. Jiang, S. Belongie, S.-N. Lim, and C. De Sa, "Differentiating through the Fréchet mean," in *ICML*, 2020.
- [36] V. Arsigny, P. Fillard, X. Pennec, and N. Ayache, "Fast and simple computations on tensors with log-Euclidean metrics." Ph.D. dissertation, INRIA, 2005.
- [37] Z. Lin, "Riemannian geometry of symmetric positive definite matrices via Cholesky decomposition," *SIMAX*, 2019.
- [38] N. Boumal and P.-A. Absil, "A discrete regression method on manifolds and its application to data on $so(n)$," *IFAC Proceedings Volumes*, vol. 44, no. 1, pp. 2284–2289, 2011.
- [39] Y. Thanwerdas, "Permutation-invariant log-Euclidean geometries on full-rank correlation matrices," *SIMAX*, 2024.
- [40] M. P. Do Carmo and J. Flaherty Francis, *Riemannian Geometry*. Springer, 1992, vol. 6.
- [41] Z. Chen, Y. Song, Y. Liu, and N. Sebe, "A Lie group approach to Riemannian batch normalization," in *ICLR*, 2024.
- [42] L. W. Tu, *An introduction to manifolds*. Springer, 2011.
- [43] J. M. Lee, *Introduction to Riemannian manifolds*. Springer, 2018, vol. 2.
- [44] B. C. Hall and B. C. Hall, *Lie groups, Lie algebras, and representations*. Springer, 2013.
- [45] M. Fréchet, "Les éléments aléatoires de nature quelconque dans un espace distancié," in *Annales de l'institut Henri Poincaré*, vol. 10, no. 4, 1948, pp. 215–310.
- [46] B. Afsari, "Riemannian L_p center of mass: Existence, uniqueness, and convexity," *Proceedings of the American Mathematical Society*, vol. 139, no. 2, pp. 655–673, 2011.
- [47] Q. Wang, J. Xie, W. Zuo, L. Zhang, and P. Li, "Deep CNNs meet global covariance pooling: better representation and generalization," *IEEE TPAMI*, 2020.
- [48] F. López, B. Pozzetti, S. Trettel, M. Strube, and A. Wienhard, "Vector-valued distance and Gyrocalculus on the space of symmetric positive definite matrices," in *NeurIPS*, 2021.
- [49] Z. Chen, T. Xu, X.-J. Wu, R. Wang, and J. Kittler, "Hybrid Riemannian graph-embedding metric learning for image set classification," *IEEE TBD*, 2021.
- [50] X. Pennec, P. Fillard, and N. Ayache, "A Riemannian framework for tensor computing," *IJCV*, 2006.
- [51] Y. Thanwerdas and X. Pennec, "O(n)-invariant Riemannian metrics on SPD matrices," *Linear Algebra and its Applications*, vol. 661, pp. 163–201, 2023.
- [52] M. Berger, *A panoramic view of Riemannian geometry*. Springer, 2003.
- [53] H. Karcher, "Riemannian center of mass and mollifier smoothing," *Communications on Pure and Applied Mathematics*, vol. 30, no. 5, pp. 509–541, 1977.
- [54] J. H. Manton, "A globally convergent numerical algorithm for computing the centre of mass on compact Lie groups," in *The 8th Control, Automation, Robotics and Vision Conference, 2004.*, vol. 3. IEEE, 2004, pp. 2211–2216.
- [55] P. David and W. Gu, "A Riemannian structure for correlation matrices," *Operators and Matrices*, vol. 13, no. 3, pp. 607–627, 2019.
- [56] I. Archakov and P. R. Hansen, "A new parametrization of correlation matrices," *Econometrica*, vol. 89, no. 4, pp. 1699–1715, 2021.
- [57] R. J. Kobler, J.-i. Hirayama, and M. Kawanabe, "Controlling the Fréchet variance improves batch normalization on the symmetric positive definite manifold," in *ICASSP*, 2022.
- [58] X. Pennec, "Probabilities and statistics on Riemannian manifolds: A geometric approach," Ph.D. dissertation, INRIA, 2004.
- [59] Y. Wang and G. S. Chirikjian, "Error propagation on the euclidean group with applications to manipulator kinematics," *IEEE Transactions on Robotics*, vol. 22, no. 4, pp. 591–602, 2006.
- [60] R. Chakraborty and B. C. Vemuri, "Statistics on the Stiefel manifold: theory and applications," *The Annals of Statistics*, vol. 47, no. 1, pp. 415–438, 2019.
- [61] F. Barbaresco, "Gaussian distributions on the space of symmetric positive definite matrices from Souriau's Gibbs state for Siegel domains by coadjoint orbit and moment map," in *Geometric Science of Information: 5th International Conference*, 2021.
- [62] S. Said, L. Bombrun, Y. Berthoumieu, and J. H. Manton, "Riemannian Gaussian distributions on the space of symmetric positive definite matrices," *IEEE TIT*, 2017.
- [63] Y. Thanwerdas and X. Pennec, "Exploration of balanced metrics on symmetric positive definite matrices," in *Geometric Science of Information: 4th International Conference, GSI 2019, Toulouse, France, August 27–29, 2019, Proceedings 4*. Springer, 2019, pp. 484–493.

- [64] —, “Is affine-invariance well defined on SPD matrices? a principled continuum of metrics,” in *Geometric Science of Information: 4th International Conference*, 2019.
- [65] G. Bédigneul and O.-E. Ganea, “Riemannian adaptive optimization methods,” in *ICLR*, 2019.
- [66] M. Müller, T. Röder, M. Clausen, B. Eberhardt, B. Krüger, and A. Weber, “Documentation mocap database HDM05,” Universität Bonn, Technical Report, 2007.
- [67] G. Garcia-Hernando, S. Yuan, S. Baek, and T.-K. Kim, “First-person hand action benchmark with RGB-D videos and 3D hand pose annotations,” in *CVPR*, 2018.
- [68] M. F. Hinss, L. Darnet, B. Somon, E. Jahanpour, F. Lotte, S. Ladouce, and R. N. Roy, “An EEG dataset for cross-session mental workload estimation: Passive BCI competition of the Neuroergonomics Conference 2021,” 2021.
- [69] Z. Chen, Y. Song, T. Xu, Z. Huang, X.-J. Wu, and N. Sebe, “Adaptive Log-Euclidean metrics for SPD matrix learning,” *IEEE TIP*, 2024.
- [70] T. de Surrél, S. Chevallier, F. Lotte, and F. Yger, “Geometry-aware visualization of high dimensional symmetric positive definite matrices,” *TMLR*, 2025.
- [71] V. Bloom, D. Makris, and V. Argyriou, “G3D: A gaming action dataset and real time action recognition evaluation framework,” in *CVPR Workshops*, 2012.
- [72] A. Shahroudy, J. Liu, T.-T. Ng, and G. Wang, “NTU RGB+ D: A large scale dataset for 3D human activity analysis,” in *CVPR*, 2016.
- [73] M. Lezcano Casado, “Trivializations for gradient-based optimization on manifolds,” *NeurIPS*, 2019.
- [74] W. S. Kendall, “Probability, convexity, and harmonic maps with small image i: uniqueness and fine existence,” *Proceedings of the London Mathematical Society*, vol. 3, no. 2, pp. 371–406, 1990.
- [75] I. Chavel, *Riemannian geometry: a modern introduction*. Cambridge university press, 1995, no. 108.
- [76] D. Groisser, “Newton’s method, zeroes of vector fields, and the riemannian center of mass,” *Advances in Applied Mathematics*, vol. 33, no. 1, pp. 95–135, 2004.
- [77] R. Bhatia, *Positive Definite Matrices*. Princeton University Press, 2009.
- [78] —, *Matrix analysis*. Springer Science & Business Media, 2013, vol. 169.
- [79] Y. Thanwerdas, “Riemannian and stratified geometries on covariance and correlation matrices,” Ph.D. dissertation, Université Côte d’Azur, 2022.
- [80] X. Pennec and N. Ayache, “Uniform distribution, distance and expectation problems for geometric features processing,” *Journal of Mathematical Imaging and Vision*, vol. 9, pp. 49–67, 1998.
- [81] H. Yong, J. Huang, D. Meng, X. Hua, and L. Zhang, “Momentum batch normalization for deep learning with small batch size,” in *ECCV*, 2020.
- [82] C. Ionescu, O. Vantzos, and C. Sminchisescu, “Matrix backpropagation for deep networks with structured layers,” in *ICCV*, 2015.
- [83] I. Murray, “Differentiation of the Cholesky decomposition,” *arXiv preprint arXiv:1602.07527*, 2016.
- [84] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimselshein, L. Antiga *et al.*, “Pytorch: An imperative style, high-performance deep learning library,” in *NeurIPS*, 2019.
- [85] R. Wang, X.-J. Wu, and J. Kittler, “SymNet: A simple symmetric positive definite manifold deep learning method for image set classification,” *IEEE TNNLS*, 2021.
- [86] V. Jayaram and A. Barachant, “MOABB: trustworthy algorithm benchmarking for BCIs,” *Journal of Neural Engineering*, vol. 15, no. 6, p. 066011, 2018.
- [87] A. Gramfort, “MEG and EEG data analysis with MNE-Python,” *Frontiers in Neuroscience*, vol. 7, 2013.
- [88] R. Vemulapalli, F. Arrate, and R. Chellappa, “Human action recognition by representing 3D skeletons as points in a Lie group,” in *CVPR*, 2014.
- [89] T. de Surrél, F. Lotte, S. Chevallier, and F. Yger, “Wrapped gaussian on the manifold of symmetric positive definite matrices,” *arXiv preprint arXiv:2502.01512*, 2025.
- [90] Y. Thanwerdas and X. Pennec, “The geometry of mixed-Euclidean metrics on symmetric positive definite matrices,” *Differential Geometry and its Applications*, vol. 81, p. 101867, 2022.
- [91] E. Zacur, M. Bossa, and S. Olmos, “Left-invariant Riemannian geodesics on spatial transformation groups,” *SIMAX*, 2014.



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LieBN: Batch Normalization over Lie Groups (Supplement)

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APPENDIX A

NOTATIONS

For better clarity, we summarize all the notations used in this paper in Tab. 12.

TABLE 12: Summary of notations.

Notation	Explanation
$\{\mathcal{M}, \odot, g\}$ or abbreviated as $\mathcal{M}$	Lie group with a group operation $\odot$ and an invariant metric g
g^L and g^R	Left-invariant and right-invariant metrics
$P_{\odot}^{-1}$	Group inverse of $P \in \mathcal{M}$
$T_P \mathcal{M}$	Tangent space at $P \in \mathcal{M}$
$g_P(\cdot, \cdot)$ or $\langle \cdot, \cdot \rangle_P$	Riemannian metric at $P \in \mathcal{M}$
$\ \cdot\ _P$	Norm induced by $\langle \cdot, \cdot \rangle_P$ on $T_P \mathcal{M}$
$d(\cdot, \cdot)$	Geodesic distance
FM and WFM	Fréchet mean and weighted Fréchet mean
Exp_P and Log_P	Riemannian exponentiation and logarithm at P
$\gamma_{(P,Q)}(t)$	Geodesic connecting P and Q
$\text{PT}_{P \rightarrow Q}$	Riemannian parallel transportation along the geodesic connecting P and Q
L_P and R_P	Lie group left and right translation by $P \in \mathcal{M}$
$f_{*,P}$	Differential map of the smooth map f at $P \in \mathcal{M}$
f^*g	Pullback metric by f from g
$\mathbb{R}^{n \times n}$	Euclidean space of $n \times n$ real matrices
$\mathcal{S}_{++}^n$	SPD manifold of $n \times n$ SPD matrices
$\mathcal{S}_{++}^n$	Euclidean space of $n \times n$ symmetric matrices
LT^n	Euclidean space of $n \times n$ lower triangular matrices
$\text{SO}(n)$	Lie group of $n \times n$ rotation matrices
$\mathfrak{so}(n)$	Euclidean space of $n \times n$ skew-symmetric matrices
$\langle \cdot, \cdot \rangle$ and $\ \cdot\ _F$	Standard Frobenius inner product and the induced norm
$\langle \cdot, \cdot \rangle^{(\alpha, \beta)}$ and $\ \cdot\ ^{(\alpha, \beta)}$	$O(n)$ -invariant Euclidean inner product and the induced norm
$\mathbf{ST}$	$\mathbf{ST} = \{(\alpha, \beta) \in \mathbb{R}^2 \mid \min(\alpha, \alpha + n\beta) > 0\}$
$g^{(\alpha, \beta)\text{-AI}}, g^{(\theta, \alpha, \beta)\text{-AI}}, g^{\text{CRI}}$ and $g^{\theta\text{-CRIM}}$	Riemannian metric tensor of (α, β) -AIM, (θ, α, β) -AIM, CRIM, and θ -CRIM
$\odot^{\text{AI}}, \odot^{\text{LE}}, \text{ and } \odot^{\text{LC}}$	Group operations w.r.t. AIM, LEM, and LCM
$(\cdot)_{\text{Sym}}$	$(X)_{\text{Sym}} = X + X^T, \forall X \in \mathbb{R}^{n \times n}$
$(\cdot)_{\frac{1}{2}}$	$(X)_{\frac{1}{2}} = \lfloor X \rfloor + \frac{1}{2} \mathbb{D}(X), \forall X \in \mathbb{R}^{n \times n}$
$\mathcal{N}(M, \sigma^2)$	Riemannian Gaussian distribution
log and exp	Matrix logarithm and exponentiation
Chol	Cholesky decomposition
Dlog	Diagonal element-wise logarithm
ψ_{LC}	$\text{Dlog} \circ \text{Chol}$
$\lfloor \cdot \rfloor$	Strictly lower triangular part of a square matrix
$P_{\theta}(\cdot)$ or $(\cdot)^{\theta}$	Matrix power function
$\mathbb{D}(\cdot)$	Returning a diagonal matrix with diagonal elements from a square matrix
$\text{diag}(\cdot)$	Returns a diagonal matrix from an input vector
Hol^n and Row_0^n	Subspace of $\mathcal{S}^n$ with null diagonals and null row sum
Row_1^n	Manifold of $n \times n$ SPD matrices with unit row sum.
$\mathcal{C}_{++}^n$	Manifold of $n \times n$ full-rank correlation matrices
LT_1^n and LT_0^n	Euclidean subspaces of LT^n with unit diagonals and null diagonals
LT_{++}^n	Cholesky manifold of $n \times n$ lower triangular matrices with positive diagonals
$\otimes$	Hadamard product
Cor	$\text{Cor} : \Sigma \in \mathcal{S}_{++}^n \mapsto \mathbb{D}(\Sigma)^{-1/2} \Sigma \mathbb{D}(\Sigma)^{-1/2} \in \mathcal{C}_{++}^n$
Θ	$\Theta : C \in \mathcal{C}_{++}^n \mapsto \mathbb{D}(\text{Chol}(C))^{-1} \text{Chol}(C) \in \text{LT}_1^n$
Off	Returns a matrix in Hol^n consisting of off-diagonal elements
Log° & Exp°	Off-log and its inverse
Log^* & Exp^*	Log-scaled and its inverse
I or I_n & $\mathbf{0}$	Identity matrix & zero matrix
$\langle \cdot, \cdot \rangle^{(\alpha, \beta, \gamma)}$ and $\langle \cdot, \cdot \rangle^{(\alpha, \delta, \zeta)}$	Permutation-invariant inner products over Hol^n and Row_0^n

APPENDIX B

RIEMANNIAN STRUCTURES ON THE INVOLVED MATRIX MANIFOLDS

B.1 Existence and Uniqueness of the Weighted Fréchet Mean

Let $(\mathcal{M}, g)$ be an orientable complete Riemannian manifold equipped with a Riemannian metric g . The induced distance is denoted as $d(\cdot, \cdot)$. We denote the supremum of the sectional curvatures of $\mathcal{M}$ by Δ . We recover the theorem on the existence and uniqueness of the Weighted Fréchet Mean (WFM) [46]. We acknowledge that Chakraborty *et al* [14, App. A] has also provided a summary of the following discussions.

Definition B.1 (Geodesic Ball). Let $P \in \mathcal{M}$ and $r > 0$. Then $B_r(P) = \{Q \in \mathcal{M} \mid d(P, Q) < r\}$ is the open geodesic ball at P of radius r .

Definition B.2 (Injectivity Radius [54]). The local injectivity radius at $P \in \mathcal{M}$, $r_{\text{inj}}(P)$, is the largest radius r for which $\text{Exp}_P : T_P\mathcal{M} \supset B_r(0) \rightarrow \mathcal{M}$ is a diffeomorphism onto its image. The injectivity radius of $\mathcal{M}$ is defined as $r_{\text{inj}}(\mathcal{M}) = \inf_{P \in \mathcal{M}} \{r_{\text{inj}}(P)\}$.

Within the local injectivity radius, the exponential map is invertible and we call the inverse map as Riemannian logarithmic map, $\text{Log}_P : B_{r_{\text{inj}}(P)}(P) \rightarrow B_{r_{\text{inj}}(P)}(0) \subset T_P\mathcal{M}$.

Definition B.3 (Regular Geodesic Ball [74]). An open geodesic ball $B_r(P)$ is a regular geodesic ball if $r < r_{\text{inj}}(P)$ and $r < \pi/(2\sqrt{\Delta})$, where $1/\sqrt{\Delta}$ is interpreted as ∞ for $\Delta \leq 0$.

If P, Q in a regular geodesic ball $B_r(P)$, there exists a unique geodesic $\gamma_{(P,Q)} : [0, 1] \rightarrow B_r(P)$ with $\gamma(0) = P$ and $\gamma(1) = Q$ [74].

Definition B.4 (Strong Convexity [75]). A subset $U \subset \mathcal{M}$ is strongly convex if for all $P, Q \in U$, there exists a unique length-minimizing geodesic segment between P and Q , and the geodesic segment lies entirely in U .

Definition B.5 (Convexity Radius [76]). The local convexity radius at $P \in \mathcal{M}$, $r_{\text{cvx}}(P)$, is defined as

$$r_{\text{cvx}}(P) = \sup\{r \leq r_{\text{inj}}(P) \mid B_r(P) \text{ is strongly convex}\}. \quad (30)$$

The convexity radius of $\mathcal{M}$ is defined as $r_{\text{cvx}}(\mathcal{M}) = \inf_{P \in \mathcal{M}} \{r_{\text{cvx}}(P)\}$.

Theorem B.6 (Existence and Uniqueness of WFM [46]). *The WFM exists and is unique inside a geodesic ball of radius $r_{\text{cvx}}(\mathcal{M})$.*

In the main paper, we always assume the involved Exp , Log , and WFM are well-defined.

B.2 Symmetric Matrix Functions

This subsection reviews the eigenvalue function over symmetric matrices. For more in-depth discussions, please refer to [77, Ch. 2.7.13] or [78, Ch. V.3].

We denote $\mathcal{S}^n$ as the Euclidean space of $n \times n$ real symmetric matrices, and $\mathcal{S}_{++}^n$ as the SPD manifold of $n \times n$ SPD matrices. Let $\dot{I}$ be an open interval of $\mathbb{R}$ and $f : \dot{I} \rightarrow \mathbb{R}$ be a smooth function. The smooth map induced by f for any symmetric matrix S with all eigenvalues in $\dot{I}$ is defined as

$$f : S \mapsto Uf(\Sigma)U^\top \in \mathcal{S}^n, \text{ with } S = U\Sigma U^\top \text{ as the eigendecomposition.} \quad (31)$$

Its differential is known as the Daleckii-Kreĭn formula:

$$f_{*,S}(V) = U \left(L \otimes (U^\top V U) \right) U^\top, \quad \forall V \in \mathcal{S}^n, \quad (32)$$

$$L_{i,j} = \begin{cases} \frac{f(\sigma_i) - f(\sigma_j)}{\sigma_i - \sigma_j}, & \text{if } \sigma_i \neq \sigma_j \\ f'(\delta_i), & \text{otherwise} \end{cases} \quad (33)$$

where L is called the Loewner matrix with the (i, j) -th element defined as Eq. (33), and $\odot$ denotes the Hadamard product. Two special cases are the matrix logarithm: $\log : \mathcal{S}_{++}^n \rightarrow \mathcal{S}^n$ and its inverse, the matrix exponentiation $\exp : \mathcal{S}^n \rightarrow \mathcal{S}_{++}^n$.

B.3 SPD Geometries

Tab. 13 summarizes the Lie groups and invariant metric on the SPD manifold with the following notations. Let $P, Q \in \mathcal{S}_{++}^n$ be SPD matrices and $V, W \in T_P\mathcal{S}_{++}^n$ be tangent vectors. We denote the matrix logarithm, exponentiation, and Cholesky decomposition by $\log(\cdot)$, $\exp(\cdot)$, and $\text{Chol}(\cdot)$, respectively. The differentials at P are $\log_{*,P}$ and $\text{Chol}_{*,P}$. The Cholesky factors of P and Q are denoted as $L = \text{Chol}(P)$ and $K = \text{Chol}(Q)$. The corresponding tangent vectors are $X = \text{Chol}_{*,P}(V)$ and $Y = \text{Chol}_{*,P}(W)$ for LCM. $\mathbb{K}$, $\mathbb{L}$, $\mathbb{X}$, and $\mathbb{Y}$ are diagonal matrices with diagonal elements from K , L , X , and Y , respectively. $[\cdot]$ is the strictly lower part of a square matrix. The norms induced by $\langle \cdot, \cdot \rangle^{(\alpha, \beta)}$ and the standard Frobenius norm (induced by the standard inner product $\langle \cdot, \cdot \rangle$) by $\|\cdot\|^{(\alpha, \beta)}$ and $\|\cdot\|_F$, respectively.

B.4 Rotation Geometries

Given $R, S \in \text{SO}(n)$, and tangent vectors $A_1, A_2 \in T_R\text{SO}(n)$, Tab. 14 summarizes all the associated operators on $\text{SO}(n)$ under the invariant metric, where $\langle \cdot, \cdot \rangle$ denotes the standard Frobenius inner product.

B.5 Correlation Geometries

The involved four geometries in Sec. 2.4 on the correlation matrices can be classified into two classes: (1) non-permutation-invariant metrics, including ECM and LECM; and (2) permutation-invariant metrics, including OLM and LSM. Tab. 15 summarizes the diffeomorphisms and prototype spaces discussed in Sec. 2.4.

TABLE 13: Lie group structures and the associated Riemannian operators on SPD manifolds.

Metric	(α, β) -LEM	(α, β) -AIM	LCM
$Q \odot P$	$\exp(\log(P) + \log(Q))$	$KPK^\top$	$\text{Chol}^{-1}([L + K] + \mathbb{K}\mathbb{L})$
$g_P(V, W)$	$\langle \log_{*,P}(V), \log_{*,P}(W) \rangle^{(\alpha,\beta)}$	$\langle P^{-1}V, WP^{-1} \rangle^{(\alpha,\beta)}$	$\langle [X], [Y] \rangle + \langle \mathbb{X}\mathbb{L}^{-1}, \mathbb{Y}\mathbb{L}^{-1} \rangle$
$d(P, Q)$	$\ \log(P) - \log(Q)\ ^{(\alpha,\beta)}$	$\left\ \log \left(Q^{-\frac{1}{2}} P Q^{-\frac{1}{2}} \right) \right\ ^{(\alpha,\beta)}$	$\ \psi_{\text{LC}}(P) - \psi_{\text{LC}}(Q)\ _F$
$\text{FM}(\{P_i\})$	$\exp \left(\frac{1}{n} \sum_i \log P_i \right)$	Karcher Flow	$\psi_{\text{LC}}^{-1} \left(\frac{1}{n} \sum_i \psi_{\text{LC}}(P_i) \right)$
$\text{Log}_P Q$	$(\log_{*,P})^{-1} [\log(Q) - \log(P)]$	$P^{\frac{1}{2}} \log \left(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}} \right) P^{\frac{1}{2}}$	$(\text{Chol}^{-1})_{*,L} [[K] - [L] + \mathbb{L} \text{Dlog}(\mathbb{L}^{-1}\mathbb{K})]$
$\text{Exp}_P V$	$\exp(\log(P) + \log_{*,P}(V))$	$P^{\frac{1}{2}} \exp \left(P^{-\frac{1}{2}} V P^{-\frac{1}{2}} \right) P^{\frac{1}{2}}$	$\text{Chol}^{-1}([L] + [X] + \mathbb{L} \exp(\mathbb{X}\mathbb{L}^{-1}))$
$\gamma_{(P,Q)}(t)$	$\exp[\log(P) + t(\log(Q) - \log(P))]$	$P^{\frac{1}{2}} \left(P^{-\frac{1}{2}} Q P^{-\frac{1}{2}} \right)^t P^{\frac{1}{2}}$	$\text{Chol}^{-1} \left\{ [L] + t([K] - [L]) + \frac{\mathbb{K}^t}{\mathbb{L}^{t-1}} \right\}$
Invariance	Bi-invariance	Left-invariance	Bi-invariance
References	[36], [51], [69]	[3], [50], [51]	[21], [37], [69]

TABLE 14: Lie group structures and the associated Riemannian operators on Rotation matrices.

Operators	$R \odot S$	$g_R(A_1, A_2)$	$d(R, S)$	$\text{Log}_R S$	$\text{Exp}_R(A)$	$\gamma_{(R,S)}(t)$	FM	Invariance
Expression	RS	$\langle A_1, A_2 \rangle$	$\ \log(R^\top S)\ _F$	$R \log(R^\top S)$	$R \exp(R^\top A)$	$R \exp(t \log(R^\top S))$	[54, Alg. 1]	Bi-invariance

TABLE 15: Riemannian metrics on the correlation manifold with the associated isometric prototype spaces and diffeomorphisms.

Metric	Prototype space	Diffeomorphisms	Properties
ECM [3]	$\text{LT}_1^n = \text{LT}_0^n + I_n$	$\Theta : C \in \mathcal{C}_{++}^n \mapsto \mathbb{D}(\text{Chol}(C))^{-1} \text{Chol}(C) \in \text{LT}_1^n$ $\Theta^{-1} = \text{Cor} \circ \text{Chol}^{-1} : \text{LT}_1^n \rightarrow \mathcal{C}_{++}^n$	Bi-invariance
LECM [3]	LT_0^n	$\log \circ \Theta : \mathcal{C}_{++}^n \rightarrow \text{LT}_0^n$ $(\log \circ \Theta)^{-1} = \text{Cor} \circ \text{Chol}^{-1} \circ \exp : \text{LT}_0^n \rightarrow \mathcal{C}_{++}^n$	Bi-invariance
OLM [39]	Hol^n	$\text{Log}^\circ : C \in \mathcal{C}_{++}^n \mapsto \text{Off} \circ \log(C) \in \text{Hol}^n$ $(\text{Log}^\circ)^{-1} = \text{Exp}^\circ : H \in \text{Hol}^n \mapsto \exp(\mathcal{D}^+(H) + H) \in \mathcal{C}_{++}^n$	Bi-invariance Permutation-invariance
LSM [39]	Row_0^n	$\text{Log}^* : C \in \mathcal{C}_{++}^n \mapsto \log(\mathcal{D}^*(C)C\mathcal{D}^*(C)) \in \text{Row}_0^n$ $(\text{Log}^*)^{-1} = \text{Exp}^* : R \in \text{Row}_0^n \mapsto \text{Cor}(\exp(R)) \in \mathcal{C}_{++}^n$	Bi-invariance Permutation-invariance

B.5.1 None-permutation-invariant Metrics

The non-permutation-invariant metrics [3], namely ECM and LECM, are defined by isometries:

$$\text{ECM: } \mathcal{C}_{++}^n \xrightarrow[\Theta^{-1} = \text{Cor} \circ \text{Chol}^{-1}]{\Theta} \text{LT}_1^n = I_n + \text{LT}_0^n, \quad (34)$$

$$\text{LECM: } \mathcal{C}_{++}^n \xrightarrow[(\log \circ \Theta)^{-1} = \text{Cor} \circ \text{Chol}^{-1} \circ \exp]{\log \circ \Theta} \text{LT}_0^n, \quad (35)$$

ECM and LECM. For any $C \in \mathcal{C}_{++}^n$, $V \in T_C \mathcal{C}_{++}^n \cong \text{Hol}^n$, $K \in \text{LT}_1^n$ and $X, \xi \in \text{LT}_0^n$, the involved maps and their

TABLE 16: Riemannian operators under the non-permutation-invariant log-Euclidean Metrics. Here, $C, C' \in \mathcal{C}_{++}^n$ are correlation matrices and $V, W \in T_C \mathcal{C}_{++}^n \cong \text{Hol}^n$ are tangent vectors.

Operation	ECM	LECM
$g_C(V, W)$	$\langle \Theta_{*,C}(V), \Theta_{*,C}(W) \rangle$	$\langle (\log \circ \Theta)_{*,C}(V), (\log \circ \Theta)_{*,C}(W) \rangle$
$\text{Exp}_C(V)$	$\Theta^{-1}(\Theta(C) + \Theta_{*,C}(V))$	$(\log \circ \Theta)^{-1}(\log \circ \Theta(C) + (\log \circ \Theta)_{*,C}(V))$
$\text{Log}_C(C')$	$\Theta_{*,\Theta(C)}^{-1}(\Theta(C') - \Theta(C))$	$(\log \circ \Theta)_{*,\log \circ \Theta(C)}^{-1}(\log \circ \Theta(C') - \log \circ \Theta(C))$
$\gamma(t; C, C')$	$\Theta^{-1}((1-t)\Theta(C) + t\Theta(C'))$	$(\log \circ \Theta)^{-1}((1-t)\log \circ \Theta(C) + t\log \circ \Theta(C'))$
$d(C, C')$	$\ \Theta(C) - \Theta(C')\ _F$	$\ \log \circ \Theta(C) - \log \circ \Theta(C')\ _F$
Fréchet mean	$\Theta^{-1}\left(\frac{1}{k} \sum_{i=1}^k \Theta(C_i)\right)$	$(\log \circ \Theta)^{-1}\left(\frac{1}{k} \sum_{i=1}^k \log \circ \Theta(C_i)\right)$
Curvature	0	0
$\text{PT}_{C \rightarrow C'}(V)$	$(\Theta_{*,C'})^{-1}(\Theta_{*,C}(V))$	$((\log \circ \Theta)_{*,C'})^{-1}((\log \circ \Theta)_{*,C}(V))$

differentials in ECM and LECM are

$$\Theta(C) = \mathbb{D}(L)^{-1}L, \quad (36)$$

$$\Theta^{-1}(K) = \mathbb{D}\left(KK^\top\right)^{-\frac{1}{2}}KK^\top\mathbb{D}\left(KK^\top\right)^{-\frac{1}{2}}, \quad (37)$$

$$\log(K) = \sum_{k=1}^{n-1} \frac{(-1)^{k-1}}{k} (K - I_n)^k, \quad (38)$$

$$\exp(\xi) = \sum_{k=0}^{n-1} \frac{1}{k!} \xi^k, \quad (39)$$

$$\Theta_{*,C}(V) = \Theta(C) \left(L^{-1}VL^{-\top}\right)_{\frac{1}{2}} - \frac{1}{2}\mathbb{D}\left(L^{-1}VL^{-\top}\right)\Theta(C), \quad (40)$$

$$(\Theta_{*,C})^{-1}(\xi) = \left(L\xi^\top - C\mathbb{D}\left(L\xi^\top\right)\right)\mathbb{D}(L) + \mathbb{D}(L)\left(\xi L^\top - \mathbb{D}\left(L\xi^\top\right)C\right), \quad (41)$$

$$\log_{*,K}(\xi) = \sum_{k=1}^{n-1} \frac{(-1)^{k-1}}{k} \left[(K - I_n)^{k-1}\xi + (K - I_n)^{k-2}\xi(K - I_n) + \cdots + \xi(K - I_n)^{k-1}\right], \quad (42)$$

$$\exp_{*,X}(\xi) = \sum_{k=1}^{n-1} \frac{1}{k!} \left(X^{k-1}\xi + X^{k-2}\xi X + \cdots + \xi X^{k-1}\right), \quad (43)$$

$$(\log \circ \Theta)_{*,C}(V) = \log_{*,\Theta(C)}(\Theta_{*,C}(V)), \quad (44)$$

$$\text{Chol}_{*,C}(V) = L \left(L^{-1}VL^{-\top}\right)_{\frac{1}{2}}, \quad (45)$$

$$(\text{Chol}_{*,C})^{-1}(Z) = LZ^\top + ZL^\top, \quad \forall Z \in T_L \text{LT}_{++}^n \cong \text{LT}^n. \quad (46)$$

where L is the Cholesky factor of C , I_n is the $n \times n$ identity matrix and LT_{++}^n is the Cholesky manifold of $n \times n$ Cholesky matrices. Due to the nilpotency of LT_0^n , the matrix logarithm over LT_1^n and exponentiation over LT_0^n are free from eigendecomposition. With the above equations, Tab. 16 summarizes the Riemannian operators under ECM and LECM.

B.5.2 Permutation-invariant Metrics

Let $\mathfrak{S}^n$ be the group of permutation matrices $P_\sigma = [\delta_{i,\sigma(j)}]_{1 \leq i,j \leq n}$ by permutation σ , and $\mathcal{D}^\pm(n) = \{\text{diag}((\varepsilon_1, \dots, \varepsilon_n)), \varepsilon \in \{-1, 1\}^n\}$ be the group of diagonal matrices with coefficients in $\{-1, 1\}$. [39, Thm. 1.1] showed that the biggest congruence action on full-rank correlation matrices is the action of signed permutation matrices:

$$\star : (A, C) \in \mathfrak{S}^\pm(n) \times \mathcal{C}_{++}^n \mapsto ACA^\top \in \mathcal{C}_{++}^n, \quad (47)$$

with $\mathfrak{S}^\pm(n) = \mathcal{D}^\pm(n)\mathfrak{S}^n$. Based on this finding, [39] proposed two permutation-invariant metrics, namely OLM and LSM, by pulling back permutation-invariant inner products via the following permutation-equivariant diffeomorphisms:

$$\mathcal{C}_{++}^n \xrightarrow[\text{Exp}^\circ]{\text{Log}^\circ = \text{Off} \circ \text{log}} \text{Hol}^n, \quad (48)$$

$$\mathcal{C}_{++}^n \xrightarrow[\text{Exp}^\star = \text{Cor} \circ \exp]{\text{Log}^\star} \text{Row}_0^n, \quad (49)$$

$$\text{Exp}^\circ : \text{Hol}^n \ni H \mapsto \exp(\mathcal{D}^+(H) + H), \quad (50)$$

$$\text{Log}^\star : \mathcal{C}_{++}^n \ni C \mapsto \log(\mathcal{D}^\star(C)C\mathcal{D}^\star(C)) \in \text{Row}_0^n. \quad (51)$$

where $\log(\cdot)$ and $\exp(\cdot)$ are symmetric matrix logarithm and exponentiation. The involved $\mathcal{D}^+$ and $\mathcal{D}^\star$ can be formally expressed as $\mathcal{D}^+ : \text{Hol}^n \rightarrow \mathbb{D}^n$ and $\mathcal{D}^\star : \mathcal{C}_{++}^n \rightarrow \mathbb{D}_+^n$, where $\mathbb{D}^n$ denotes the Euclidean space of $n \times n$ diagonal matrices, and $\mathbb{D}_+^n$ is the submanifold of $\mathbb{D}^n$, consisting of positive diagonal matrices.

TABLE 17: Riemannian geometries under the permutation-invariant log-Euclidean Metrics.

Operation	OLM	LSM
$g_C(V, W)$	$\langle \text{Log}_{*,C}^\circ(V), \text{Log}_{*,C}^\circ(W) \rangle^{(\alpha, \beta, \gamma)}$	$\langle \text{Log}_{*,C}^*(V), \text{Log}_{*,C}^*(W) \rangle^{(\alpha, \delta, \zeta)}$
$\text{Exp}_C(V)$	$\text{Exp}^\circ(\text{Log}^\circ(C) + \text{Log}_{*,C}^\circ(V))$	$\text{Exp}^*(\text{Log}^*(C) + \text{Log}_{*,C}^*(V))$
$\text{Log}_C(C')$	$\text{Exp}_{*, \text{Log}^\circ(C)}^\circ(\text{Log}^\circ(C') - \text{Log}^\circ(C))$	$\text{Exp}_{*, \text{Log}^*(C)}^*(\text{Log}^*(C') - \text{Log}^*(C))$
$\gamma(t; C, C')$	$\text{Exp}^\circ((1-t)\text{Log}^\circ(C) + t\text{Log}^\circ(C'))$	$\text{Exp}^*((1-t)\text{Log}^*(C) + t\text{Log}^*(C'))$
$d(C, C')$	$\ \text{Log}^\circ(C) - \text{Log}^\circ(C')\ ^{(\alpha, \beta, \gamma)}$	$\ \text{Log}^*(C) - \text{Log}^*(C')\ ^{(\alpha, \delta, \zeta)}$
Fréchet mean	$\text{Exp}^\circ\left(\frac{1}{k} \sum_{i=1}^k \text{Log}^\circ(C_i)\right)$	$\text{Exp}^*\left(\frac{1}{k} \sum_{i=1}^k \text{Log}^*(C_i)\right)$
Curvature	0	0
$\text{PT}_{C \rightarrow C'}(V)$	$\left(\text{Log}_{*,C'}^\circ\right)^{-1} \left(\text{Log}_{*,C}^\circ(V)\right)$	$\left(\text{Log}_{*,C'}^*\right)^{-1} \left(\text{Log}_{*,C}^*(V)\right)$
Invariance	Bi-invariance Permutation-invariance Signed-permutation-invariance ($\beta = \gamma = 0$)	Bi-invariance Permutation-invariance

The differentials of Log° and Log^* and their inverses can be calculated by the differential of symmetric matrix logarithm and exponentiation [39, Thms. 2.4 and 4.1]. Given $C \in \mathcal{C}_{++}^n$, tangent vector $V \in T_C \mathcal{C}_{++}^n \cong \text{Hol}^n$, $H, W \in \text{Hol}^n$, and $S = H + \mathcal{D}^+(H) = U\Delta U^\top$, the differential of Log° and its inverse Exp° are

$$\text{Log}_{*,C}^\circ(V) = \text{Off}(\log_{*,C}(V)), \quad (52)$$

$$\text{Exp}_{*,H}^\circ(W) = \exp_{*,S}\left(W + \mathcal{D}_{*,H}^+(W)\right), \quad (53)$$

$$\mathcal{D}_{*,H}^+(W) = -\text{diag}\left((H^0)^{-1} \mathbb{D}(\exp_{*,S}(W)) \mathbf{1}\right), \quad (54)$$

$$\mathcal{S}_{++}^n \ni H_{il}^0 = \sum_{j,k} P_{ij} P_{ik} P_{lj} P_{lk} L_{j,k}, \quad (55)$$

where L is the Loewner matrix of $\exp_{*,S}$, and $\mathbf{1}$ is the vector of all 1 entities. Here, $\log_*$ and $\exp_*$ can be calculated using the Daleckii-Krein formula of the symmetric matrix, while $\text{diag}(\cdot) : \mathbb{R}^n \rightarrow \mathbb{D}^n$ returns a diagonal matrix from an input vector. Further denoting $X, Y \in \text{Row}_0(n)$ and $\Sigma = \mathcal{D}^*(C)C\mathcal{D}^*(C)$, the differentials of Log^* and its inverse Exp^* are

$$\text{Log}_{*,C}^*(V) = \log_{*,\Sigma}\left(\Delta V \Delta + \frac{1}{2}(V^0 \Sigma + \Sigma V^0)\right), \quad (56)$$

$$\text{Exp}_{*,X}^*(Y) = \Delta^{-1} \left[\exp_{*,X}(Y) - \frac{1}{2}(\Delta^{-2} \mathbb{D}(\exp_{*,X}(Y)) \Sigma + \Sigma \mathbb{D}(\exp_{*,X}(Y)) \Delta^{-2}) \right] \Delta^{-1} \quad (57)$$

with $\Delta = \mathbb{D}(\Sigma)^{1/2}$ and $V^0 = -2 \text{diag}\left((I_n + \Sigma)^{-1} \Delta V \Delta \mathbf{1}\right)$.

As both Log_*^* and Log_*° are permutation-equivariant [39], permutation-invariant metrics over the correlation manifold can be induced by permutation-invariant inner products over Hol^n and Row_0^n , respectively. The following two theorems review such inner products.

Theorem B.7 (Permutation-Invariant Inner Products on Hol^n [79]). *Supposing $n \geq 4$, permutation-invariant inner products on Hol^n are:*

$$\langle X_1, X_2 \rangle^{(\alpha, \beta, \gamma)} = \alpha \text{tr}(X_1 X_2) + \beta \text{Sum}(X_1 X_2) + \gamma \text{Sum}(X_1) \text{Sum}(X_2), \quad \forall X_1, X_2 \in \text{Hol}^n, \quad (58)$$

with $\alpha > 0$, $2\alpha + (n-2)\beta > 0$, and $\alpha + (n-1)(\beta + n\gamma) > 0$. For $n = 3$, permutation-invariant inner products have the same form with $\alpha = 0$:

$$\langle X_1, X_2 \rangle^{(\alpha, \beta, \gamma)} = \beta \text{Sum}(X_1 X_2) + \gamma \text{Sum}(X_1) \text{Sum}(X_2), \quad \text{with } \beta > 0 \text{ and } \beta + 3\gamma > 0. \quad (59)$$

For $n = 2$, they have the same form with $\alpha = \beta = 0$:

$$\langle X_1, X_2 \rangle^{(\alpha, \beta, \gamma)} = \gamma \text{Sum}(X_1) \text{Sum}(X_2), \quad \text{with } \gamma > 0. \quad (60)$$

Theorem B.8 (Permutation-Invariant Inner Products on Row_0^n [39]). *For $n \geq 4$, permutation-invariant inner products on Row_0^n are*

$$\langle Y_1, Y_2 \rangle^{(\alpha, \delta, \zeta)} = \alpha \text{tr}(Y_1 Y_2) + \delta \text{tr}(\mathbb{D}(Y_1) \mathbb{D}(Y_2)) + \zeta \text{tr}(Y_1) \text{tr}(Y_2), \quad \forall Y_1, Y_2 \in \text{Row}_0^n, \quad (61)$$

with $\alpha > 0$, $n\alpha + (n-2)\delta > 0$, and $n\alpha + (n-1)(\delta + n\zeta) > 0$. For $n = 3$, the permutation-invariant inner products have the same form with $\alpha = 0$. For $n = 2$, they have the same form with $\alpha = \delta = 0$.

As shown by [79], the OLM is further signed-permutation invariant under $\beta = \gamma = 0$, reduced to the scaled canonical Euclidean inner product:

$$\langle V, W \rangle^{(\alpha, \beta, \gamma)} = \alpha \langle V, W \rangle, \quad \forall V, W \in \text{Hol}^n. \quad (62)$$

In the main paper, we assume that $\langle \cdot, \cdot \rangle^{(\alpha, \beta, \gamma)}$ and $\langle \cdot, \cdot \rangle^{(\alpha, \delta, \zeta)}$ are the canonical Euclidean inner product. Tab. 17 summarizes the Riemannian structures of OLM and LSM.

Remark B.9. $\mathcal{D}^+$ is also well-defined over $\mathcal{S}^n$, a surjective map $\mathcal{D}^+ : \mathcal{S}^n \rightarrow \mathbb{D}^n$. In this way, $\text{Exp}^\circ : \mathcal{S}^n \rightarrow \mathcal{C}_{++}^n$ is no longer bijective [39, Thm. 2.1]. Similarly, $\mathcal{D}^*$ is well defined over $\mathcal{S}_{++}^n$, a surjective map $\mathcal{D}^* : \mathcal{S}_{++}^n \rightarrow \mathbb{D}_+^n$. Consequently, $\text{Log}^* : \mathcal{S}_{++}^n \rightarrow \text{Row}_0^n$ is no longer bijective [39, Thm. 3.5].

APPENDIX C

BASIC LAYERS IN SPDNET AND TSMNET

SPDNet [9] is the most classic SPD neural network. SPDNet mimics the conventional densely connected feedforward network, consisting of three basic building blocks

$$\text{BiMap layer: } S^k = W^k S^{k-1} W^{k\top}, \text{ with } W^k \text{ semi-orthogonal,} \quad (63)$$

$$\text{ReEig layer: } S^k = U^{k-1} \max(\Sigma^{k-1}, \epsilon I_n) U^{k-1\top}, \text{ with } S^{k-1} = U^{k-1} \Sigma^{k-1} U^{k-1\top}, \quad (64)$$

$$\text{LogEig layer: } S^k = \log(S^{k-1}). \quad (65)$$

where $\max(\cdot)$ is element-wise maximization. BiMap and ReEig mimic transformation and non-linear activation, while LogEig maps SPD matrices into the tangent space at the identity matrix for classification.

TSMNet [33] can be illustrate as $f_{tc} \rightarrow f_{sc} \rightarrow f_{BiMap} \rightarrow f_{ReEig} \rightarrow f_{LogEig}$, where f_{tc} and f_{sc} denote temporal and spatial convolution, respectively.

APPENDIX D

STATISTICAL RESULTS OF SCALING IN LIEBN

In this section, we will show the effect of our scaling (Eq. (12)) on the population. We will see that while the resulting population variance is generally agnostic, it becomes analytic under certain circumstances, such as SPD manifolds under LEM or LCM. As a result, Eq. (12) can normalize and transform the latent Gaussian distribution.

To simplify, let $\phi_s(P) = \text{Exp}_E[s \text{Log}_E(P)]$. Similar to the main paper, $\mathcal{M}$ denotes a Lie group with a left-invariant metric. First, we present a lemma on the resulting P.D.F. of a random point transformed by ϕ_s .

Lemma D.1. *Given a random point X distributed over $\mathcal{M}$ with P.D.F. p_X , the P.D.F. of $Y = \phi_s(X)$ is given by:*

$$p_Y(Q) = \Delta p_X(\phi_s^{(-1)}(Q)). \quad (66)$$

where $\Delta = \frac{|\phi_{s*}^{-1}|}{|L_{\phi_s^{-1}(Q) \odot Q^{-1}*}|}$. Here $|\cdot|$ denotes the determinant, and ϕ_{s*}^{-1} and $L_{\phi_s^{-1}(Q) \odot Q^{-1}*}$ are the differentials.

Proof. For the sake of simplicity, we will denote ϕ_s as ϕ throughout this proof. The volume element w.r.t. a left-invariant metric is the Haar measure [80, Sec. 3.2]:

$$d_L \mathcal{M}(P) = \frac{dP}{|L_{P*,E}|}, \quad (67)$$

where $|L_{P*,E}|$ is the determinant⁶ of the differential of L_P at the neutral element E . Then we have

$$\begin{aligned} d_L \mathcal{M}(\phi^{-1}(Q)) &= \frac{d\phi^{-1}(Q)}{|L_{\phi^{-1}(Q)*,E}|} \\ &= |(L_{\phi^{-1}(Q) \odot Q^{-1}*} \circ L_Q)_{*,E}|^{-1} |\phi_*^{-1}| dQ \\ &= \frac{|\phi_*^{-1}|}{|L_{\phi^{-1}(Q) \odot Q^{-1}*}|} d_L \mathcal{M}(Q) \\ &= \Delta d_L \mathcal{M}(Q). \end{aligned} \quad (68)$$

The probability of $Q = \phi(P)$ in a set $\mathcal{Y} \subset \mathcal{M}$ is

$$\begin{aligned} F(\phi(P) \in \mathcal{Y}) &= F(P \in \phi^{-1}(\mathcal{Y})) \\ &= \int_{\phi^{-1}(\mathcal{Y})} p_X(P) \cdot d_L \mathcal{M}(P) \\ &= \int_{\mathcal{Y}} p_X(\phi^{-1}(Q)) d_L \mathcal{M}(\phi^{-1}(Q)) \\ &= \int_{\mathcal{Y}} \Delta p_X(\phi^{-1}(Q)) d_L \mathcal{M}(Q). \end{aligned} \quad (69)$$

6. This should be more precisely understood as the determinant of the matrix representation of $L_{P*,E}$ under a local coordinate

Therefore, the density of $Y = \phi(X)$ is

$$p_Y(Q) = \Delta p_X(\phi^{(-1)}(Q)). \quad (70)$$

□

The above lemma implies that when Δ is a constant, Y also follows a Gaussian distribution.

Corollary D.2. *Following the notations in Thm. D.1, if $\Delta = c$ is a constant and $X \sim \mathcal{N}(E, \sigma^2)$, then Y also follows a Gaussian distribution, i.e. $Y \sim \mathcal{N}(E, s^2\sigma^2)$*

Proof.

$$\begin{aligned} p_Y(Q) &= ck(\sigma) \exp\left(-\frac{d(\phi_s^{-1}(Q), E)^2}{2\sigma^2}\right) \\ &= k'(\delta) \exp\left(-\frac{d(\text{Exp}_E^{1/s} \text{Log}_E(Q), E)^2}{2\sigma^2}\right) \\ &= k'(\delta) \exp\left(-\frac{\|\text{Log}_E(Q)\|_E^2}{2s^2\sigma^2}\right) \\ &= k'(\delta) \exp\left(-\frac{d(Q, E)}{2s^2\sigma^2}\right), \end{aligned} \quad (71)$$

where $\|\cdot\|_E$ is the norm of the tangent space at the neutral element E . □

Thm. D.2 implies that when $\Delta = c$, ϕ_s can scale the population variance and further transform the Gaussian distribution. Simple computations show that in the standard Euclidean space $\mathbb{R}^n$, $\Delta = 1/s$. Therefore, it is natural to expect that the pullback of $\mathbb{R}^n$ also enjoys constant Δ .

Proposition D.3. *Consider an n -dimensional Lie group $\mathcal{M}$ pulled back from the standard Euclidean space $\mathbb{R}^n$ by the diffeomorphism $\psi : \mathcal{M} \rightarrow \mathbb{R}^n$. In other words, the group operations and Riemannian metric on $\mathcal{M}$ are defined by ψ from $\mathbb{R}^n$. Then Δ remains constant on $\mathcal{M}$.*

Proof. To simplify notation, we denote ϕ_s as ϕ . Under the given assumption, the group addition and Riemannian metric on $\mathcal{M}$ are defined as follows:

$$\begin{aligned} \forall P, Q \in \mathcal{M}, P \odot Q &= \psi^{-1}(\psi(P) + \psi(Q)) \\ g &= \psi^* g^E, \end{aligned} \quad (72)$$

where g^E is the standard Euclidean metric. Therefore, ϕ can be simplified as

$$\begin{aligned} \phi(P) &= \text{Exp}_E[s \text{Log}_E(P)] \\ &= \psi^{-1}\left(\tilde{\text{Exp}}_0\left[\psi_{*,E}\left(s\psi_{*,0}^{-1}\tilde{\text{Log}}_0\psi(P)\right)\right]\right) \\ &= \psi^{-1}(s\psi(P)), \end{aligned} \quad (73)$$

where $\tilde{\text{Exp}}$ and $\tilde{\text{Log}}$ are the Riemannian exponential and logarithmic maps in $\mathbb{R}^n$, which are reduced to vector addition and subtraction, respectively. Therefore, the inverse of ϕ is

$$\phi^{-1}(P) = \psi^{-1}(1/s\psi(P)). \quad (74)$$

Besides, $L_{\phi^{-1}(Q) \odot Q^{-1}}$ can also be further simplified:

$$L_{\phi^{-1}(Q) \odot Q^{-1}}(P) = \psi^{-1}(1/s\psi(Q) - \psi(Q) + \psi(P)) \quad (75)$$

The differentials of Eqs. (74) and (75) at Q are

$$\phi_{*,Q}^{-1} = \frac{1}{s}\psi_{*,1/s\psi(Q)}^{-1} \circ \psi_{*,Q}, \quad (76)$$

$$L_{\phi^{-1}(Q) \odot Q^{-1}*,Q} = \psi_{*,1/s\psi(Q)}^{-1} \circ \psi_{*,Q}. \quad (77)$$

Therefore, $\Delta = 1/s$ for all $Q \in \mathcal{M}$. □

By Thm. D.3, we can directly obtain the following corollary.

Corollary D.4. *Given a Lie group $\mathcal{M}$ pulled back from the Euclidean space, and a random point $X \sim \mathcal{N}(E, \sigma^2)$ over $\mathcal{M}$, $Y = \phi_s(X) \sim \mathcal{N}(E, s^2\sigma^2)$*

In machine learning, several Lie groups are derived by the pullback from the standard Euclidean space. As shown by Chen *et al* [22], (α, β) -LEM and θ -LCM are pullback metrics from the Euclidean metric. Therefore, for the Lie groups of

SPD manifolds w.r.t. (α, β) -LEM and θ -LCM, Eq. (12) can transform the Gaussian distribution. Specifically, given a random point $X \sim \mathcal{N}(M, \sigma^2)$, Eqs. (11) to (13) transform the Gaussian distribution as:

$$\mathcal{N}(M, \sigma^2) \rightarrow \mathcal{N}(E, \sigma^2) \rightarrow \mathcal{N}(E, s^2) \rightarrow \mathcal{N}(B, s^2), \quad (78)$$

where M and σ are employed to normalize X , and ϵ in Eq. (12) is omitted. The above process exactly mirrors the transformation of Gaussian distributions within the framework of standard BN [28].

Remark D.5. A similar result to our Thm. D.4 was also presented in [34, Prop. 3]. However, in his proof, the author did not account for the Haar measure and only considered the P.D.F., casting doubt on the validity of their results. Additionally, their discussion is limited to matrix Lie groups, specifically under the distance $d(P, Q) = \|\log(P^{-1}Q)\|_F$. In contrast, we rectify their proof and consider general Lie groups.

APPENDIX E

DOMAIN-SPECIFIC MOMENTUM LIEBN FOR EEG CLASSIFICATION

Algorithm 2: Momentum LieBN (MLieBN) Algorithm

Input : A batch of activations $\{P_{1...N}\}$ over the Lie group $\{\mathcal{M}, \odot, g\}$, and a small positive constant ϵ
 running mean $\bar{M}_r = E$, running variance $\bar{v}_r^2 = 1$ for training
 running mean $\tilde{M}_r = E$, running variance $\tilde{v}_r^2 = 1$ for testing
 biasing parameter $B \in \mathcal{M}$, scaling parameter $s \in \mathbb{R}/\{0\}$,
 momentum for training and testing $\gamma_{train}, \gamma \in [0, 1]$
Output : Normalized activations $\{\tilde{P}_{1...N}\}$

if training then

 Compute batch mean M_b and variance v_b^2 of $\{P_{1...N}\}$;
 $\bar{M}_r \leftarrow \text{WFM}(\{1 - \gamma_{train}, \gamma_{train}\}, \{\bar{M}_r, M_b\})$;
 $\bar{v}_r^2 \leftarrow (1 - \gamma_{train})\bar{v}_r^2 + \gamma_{train}v_b^2$;
 $\tilde{M}_r \leftarrow \text{WFM}(\{1 - \gamma, \gamma\}, \{\tilde{M}_r, M_b\})$;
 $\tilde{v}_r^2 \leftarrow (1 - \gamma)\tilde{v}_r^2 + \gamma v_b^2$;

end

if training then $M \leftarrow \bar{M}_r, v^2 \leftarrow \bar{v}_r^2$;

else $M \leftarrow \tilde{M}_r, v^2 \leftarrow \tilde{v}_r^2$;

for $i \leftarrow 1$ **to** N **do**

 Centering to the neutral element E :

if g *is left-invariant* **then** $\bar{P}_i \leftarrow L_{M_\odot^{-1}}(P_i)$;

else $\bar{P}_i \leftarrow R_{M_\odot^{-1}}(P_i)$;

 Scaling the dispersion:

$\hat{P}_i \leftarrow \text{Exp}_E \left[\frac{s}{\sqrt{v^2 + \epsilon}} \text{Log}_E(\bar{P}_i) \right]$

 Biasing towards parameter B :

if g *is left-invariant* **then** $\tilde{P}_i \leftarrow L_B(\hat{P}_i)$;

else $\tilde{P}_i \leftarrow R_B(\hat{P}_i)$;

end

Kobler *et al* [33] proposed SPD domain-specific momentum batch normalization (SPDDSMBN) as a domain adaptation approach for EEG classification. SPDDSMBN, based on Eq. (9), performed normalization of mean and variance on SPD manifolds under the specific AIM. Additionally, SPDDSMBN utilized separate momentums for updating training and testing running statistics, inspired by the work of [81]. Following [33, Alg. 1], we also present a momentum LieBN (MLieBN) in Alg. 2. Here γ is fixed and γ_{train} is defined as

$$\gamma_{train} = 1 - \rho^{\frac{1}{K-1} \max(K-k, 0)} + \rho, \text{ where } \rho = \frac{1}{\text{domains_per_batch}} \quad (79)$$

Furthermore, in line with [33], we adopt multi-channel mechanisms for domain-specific MLieBN (DSMLieBN), where each domain has its own MLieBN layer. Similar to [33], we set the biasing parameter equal to the neutral element, and the scaling factor is shared across all domains. We denote Alg. 2 as MLieBN($P_j|M, s, \epsilon, \gamma, \gamma_{train}$). Then our DSMLieBN follows

$$\text{DSMLieBN}(P_j, i) = \text{MLieBN}_i(P_j|E, s, \epsilon, \gamma, \gamma_{train}), \forall P_j \in \{P_{1...N}\}, \quad (80)$$

where i is the index of the domain. We follow the official code of SPDDSMBN⁷ to implement our DSMLieBN. In a word, the only difference between DSMLieBN and SPDDSMBN is the different way of normalization.

7. <https://github.com/rkobler/TSMNet>

Analogous to Thm. 5.6, computations for DSMLieBN under pullback metrics can also be performed by mapping, calculating, and then remapping.

APPENDIX F

BACKPROPAGATION OF MATRIX FUNCTIONS

Our implementation of LieBN on SPD and correlation manifolds involves several matrix functions. Thus, we employ matrix backpropagation (BP) [82] for gradient computation. These matrix operations can be divided into Cholesky decomposition and the functions based on Eigendecomposition.

The differentiation of the Cholesky decomposition can be found in [83, Eq. 8] or [37, Props. 4]. Besides, our homemade BP of the Cholesky decomposition yields a similar gradient to the one generated by autograd of `torch.linalg.cholesky`. Therefore, during the experiments, we use `torch.linalg.cholesky`.

The second type of matrix functions is based on Eigendecomposition, such as matrix exponential, logarithm, and power. Although torch [84] supports autograd of Eigendecomposition, it requires the computation of $\frac{1}{\delta_i - \delta_j}$ [82, Props. 1], where δ_i and δ_j denote eigenvalues. This might trigger numerical instability when δ_i approximates δ_j . Following [32], we use the Daleckii-Kreĭn formula [78, Thm. V.3.3] to calculate the BP of Eigen-based matrix functions. In detail, for a matrix function defined as $X = f(S) = Uf(\Sigma)U^\top$, with $S = U\Sigma U^\top$ as the eigendecomposition of an SPD matrix, its BP is expressed as

$$\nabla_S L = U[K \odot (U^\top (\nabla_X L) U)]U^\top. \quad (81)$$

where $\nabla_X L$ is the Euclidean gradient of the loss function L w.r.t. X . Matrix K is defined as

$$K_{ij} = \begin{cases} \frac{f(\sigma_i) - f(\sigma_j)}{\sigma_i - \sigma_j} & \text{if } \sigma_i \neq \sigma_j \\ f'(\sigma_i) & \text{otherwise} \end{cases} \quad (82)$$

where $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_d)$. Eq. (82) demonstrates the numerical stability of Daleckii-Kreĭn formula.

APPENDIX G

EXPERIMENTAL DETAILS AND ADDITIONAL DISCUSSIONS

G.1 Experimental Details and Additional Discussion on the SPD Manifold

G.1.1 Datasets and Preprocessing

The **Radar** dataset [32] contains 3,000 synthetic radar signals. Following the protocol in [32], each signal is split into windows of length 20, resulting in 3,000 covariance matrices of 20×20 equally distributed in 3 classes.

The **HDM05** dataset [66] consists of 2,273 skeleton-based motion capture sequences executed by different actors. Each frame consists of 3D coordinates of 31 joints, allowing the representation of each sequence as a 93×93 covariance matrix. In line with [32], we trim the dataset to 2086 instances scattered throughout 117 classes by removing some under-represented clips.

The **FPHA** [67] includes 1,175 skeleton-based first-person hand gesture videos of 45 different categories with 600 clips for training and 575 for testing. Following [85], we represent each sequence as a 63×63 covariance matrix.

The **Hinss2021** dataset [68] is a recently released competition dataset containing EEG signals for mental workload estimation. The dataset is employed for two tasks, inter-session and inter-subject, which are treated as domain adaptation problems. Geometry-aware methods [1], [33] have demonstrated promising performance in EEG classification. We follow [33] for data preprocessing. In detail, the python package MOABB [86] and MNE [87] are used to preprocess the datasets. The applied steps include resampling the EEG signals to 250/256 Hz, applying temporal filters to extract oscillatory EEG activity in the 4 to 36 Hz range, extracting short segments (≤ 3 s) associated with a class label, and finally obtaining 40×40 SPD covariance matrices.

G.1.2 Implementation details

We use the official code of SPDNetBN⁸ [32] and TSMNet⁹ [33] to implement our experiments on the SPDNet and TSMNet backbones. For the SPDNet architecture, we compare our LieBN with SPDNetBN [32], which applies the SPDBN (Eqs. (6) and (7)) to SPDNet. Similar to SPDNetBN, we apply our LieBN after each transformation layer (BiMap layer in App. C). In the EEG application, one of the state-of-the-art methods is TSMNet with SPD domain-specific momentum batch normalization (TSMNet+SPDDSMBN) [33], which is a domain adaptation version of [57]. For a fair comparison, we also implement a domain-specific momentum LieBN, referred to as DSMLieBN (detailed in App. E). Following [33], we apply our DSMLieBN before the LogEig layer (detailed in App. C) in TSMNet. We use the standard cross-entropy loss

8. https://proceedings.neurips.cc/paper_files/paper/2019/file/6e69ebbfad976d4637bb4b39de261bf7-Supplemental.zip

9. <https://github.com/rkobler/TSMNet>

and optimize the parameters with the Riemannian AMSGrad optimizer [65]. The network architectures are represented as $\{d_0, d_1, \dots, d_L\}$, where the dimension of the parameter in the i -th BiMap layer is $d_i \times d_{i-1}$. The experiments are conducted with a learning rate of $5e^{-3}$, batch size of 30, and training epoch of 200 on the Radar, HDM05, and FPHA datasets. For the Hinss2021 dataset, following [33], we use a learning rate of $1e^{-3}$ with a weight decay of $1e^{-4}$, a batch size of 50, and a training epoch of 50.

In line with the previous work [32], [33], we use accuracy as the scoring metric for the Radar, HDM05, and FPHA datasets, and balanced accuracy (*i.e.*, the average recall across classes) for the Hinss2021 dataset. Ten-fold experiments on the Radar, HDM05, and FPHA datasets are carried out with randomized initialization and split (split is officially fixed for the FPHA dataset), while on the Hinss2021 dataset, models are fit and evaluated with a randomized leave 5% of the sessions (inter-session) or subjects (inter-subject) out cross-validation scheme.

G.1.3 Candidate Values of Hyper-parameters

We implement the SPD LieBN and DSMLieBN induced by four standard invariant metrics, namely AIM, LEM, LCM, and CRIM, along with their deformed metrics. Therefore, our method has a maximum of three hyper-parameters, *i.e.*, (θ, α, β) . As (α, β) only affects variance calculation in the LieBN framework, we set $(\alpha, \beta) = (1, 0)$ and only tune the deformation factor θ from the candidate values of ± 0.5 , ± 1 , and ± 1.5 . We denote [Baseline]+[BN_Type]+[Metric]-[θ] as the baseline endowed with a specific LieBN, such as SPDNet+LieBN-AIM-(1) and TSMNet+DSMLieBN-LCM-(1).

G.1.4 Empirical Insights on the Hyper-parameters

Our SPD LieBN has at most three types of hyper-parameters: Riemannian metric, deformation factor θ , and $O(n)$ -invariance parameters (α, β) . The general order of importance should be Riemannian metric $> \theta > (\alpha, \beta)$.

The most significant parameter is the choice of Riemannian metric, as all the geometric properties are sourced from a metric. A safe choice would start with AIM, and then decide whether to explore other metrics further. The most important reason is the property of affine invariance of AIM, which is a natural characteristic of covariance matrices. In our experiments, the LieBN-AIM generally achieves the best performance. However, AIM is not always the best metric. As shown in Tab. 7, the best result on the HDM05 dataset is achieved by LCM-based LieBN, which improves the vanilla SPDNet by 11.71%. Therefore, when choosing Riemannian metrics on SPD manifolds, a safe choice would start with AIM and extend to other metrics. Besides, if efficiency is an important factor, one should first consider LCM, as it is the most efficient one.

The second one is the deformation factor θ . As we discussed in Sec. 5.1.1, θ interpolates between different types of metrics ($\theta = 1$ and $\theta \rightarrow 0$). Inspired by this, we select θ around its deformation boundaries (1 and 0). In this paper we roughly select θ from $\{\pm 0.5, \pm 1, \pm 1.5\}$.

The less important parameters are (α, β) . Recalling Alg. 1. and Tab. 13, (α, β) only affects the calculation of variance, which should have less effects compared with the above two parameters. Therefore, we simply set $(\alpha, \beta) = (1, 0)$ during experiments.

G.1.5 The Effect of β

Recalling Eq. (3), β controls the relative importance of the trace part against the inner product. Therefore, we set the candidate values of β as $\{1, 1/n, 1/n^2, 0, -1/n + \epsilon, -1/n^2\}$, where n is the input dimension of LieBN, and ϵ is a small positive scalar to ensure $O(n)$ -invariance, *i.e.* $(\alpha, \beta) \in \mathbf{ST}$. $1/n^2$ and $1/n$ means averaging the trace in Eq. (3), while the sign of β denotes suppressing (-), enhancing (+), or neutralizing (0) the trace.

We focus on AIM-based LieBN on the HDM05 dataset. We set $\theta = 1.5$, as it is the best deformation factor under this scenario. Other network settings remain the same as the main paper. The 10-fold average results are presented in Tab. 18. Note that on this setting, $n = 30$. As expected, β has minor effects on our LieBN.

TABLE 18: The effect of different β for AIM-based LieBN on the HDM05 dataset.

Beta	$-1/30^2$	-0.03	$1/30^2$	$1/30$	1	0
Mean $\pm$ STD	68.18 $\pm$ 0.86	68.12 $\pm$ 0.74	68.20 $\pm$ 0.85	68.18 $\pm$ 0.85	68.16 $\pm$ 0.80	68.16 $\pm$ 0.68

G.2 Implementation Details on the Rotation Matrix

G.2.1 Datasets and Preprocessing

The **G3D** dataset [71] consists of 663 sequences of 20 different gaming actions. Each sequence is recorded by 3D locations of 20 joints (*i.e.*, 19 bones). Following [10], we use the cross-subject setting, where half of the subjects are used for training, and the other half for testing, respectively.

The **HDM05** dataset [66] has been discussed in App. G.1.1.

The **NTU60** [72]. It has 56,880 sequences of 3D skeleton data classified into 60 classes, where each frame contains the 3D coordinates of 25 or 50 body joints. We focus on the cross-view protocol setting [72].

Following [10], we use the code¹⁰ of [88] to represent each skeleton sequence as a point on the Lie group $SO^{N \times T}(3)$, where N and T denote spatial and temporal dimensions. As preprocessed in [10], we set T as 100, 64, and 16 on these three datasets, respectively.

G.2.2 LieNet

The LieNet consists of three basic layers: RotMap, RotPooling, and LogMap layers. The RotMap mimics the convolutional layer, while the RotPooling extends the pooling layers to rotation matrices. The LogMap layer maps the rotation matrix into the tangent space at the identity for classification. Note that the official code of LieNet¹¹ is developed by Matlab. We use the open-sourced Pytorch code¹² to implement our experiments. To reproduce LieNet more faithfully, we made the following modifications to this Pytorch code. We re-code the LogMap and RotPooling layers to make them consistent with the official Matlab implementation. In addition, we also extend the Riemannian computation of geoopt [65] into $SO(3)$ to allow for Riemannian optimizer on $SO(3)$, which is missing in the current package. We apply our LieBN before the LogMap layer. Note that the dimension of input features in LieNet is $B \times N \times T \times 3 \times 3$. We calculate Lie group statistics along the batch and temporal dimensions ($B \times T$). We denote the LieNet models with our LieBN-Left and LieBN-Right as LieNetLieBN-Left and LieNetLieBN-Right, respectively.

G.2.3 Training Details

We find that SGD is the most effective optimizer for LieNet, and thus, we adopt it for our experiments. The learning rate is set to $1e^{-2}$. The batch sizes are 30, 30, and 256 for the G3D, HDM05, and NTU60 datasets, respectively. On the NTU60 dataset, the learning rate is reduced by a factor of 10 upon model convergence, specifically at the 5th and 25th epochs for LieNetLieBN and LieNet, respectively. For each model, we apply `torch.nn.utils.clip_grad_norm_` with `max_norm=5` to the transformation matrix in the final FC layer.

G.3 Implementation Details on the Correlation Matrix

We follow the same settings as the experiments on the SPD manifold with respect to the backbone architecture, batch size, training epoch, optimizer, and learning rate. The network architecture can be denoted as BiMap-[Power-Cov2Cor-LieBN-Cor]-LogEig, where Power denotes the matrix power and Cov2Cor is $\text{Cor}(\cdot) : \mathcal{S}_{++}^n \rightarrow \mathcal{C}_{++}^n$. The matrix powers used for each dataset are presented in Tab. 19. A single iteration is sufficient to achieve saturated network performance with respect to calculating $\mathcal{D}^+$ and $\mathcal{D}^*$ in OLM and LSM, except for $\mathcal{D}^*$ on the HDM05 dataset, which requires convergence with a maximum of 20 iterations.

TABLE 19: Matrix powers in LieBN-Cor under different metrics on each dataset.

Dataset \ Metric	Metric			
	ECM	LECM	OLM	LSM
HDM05	0.75	0.5	0.5	-0.5
FPHA	-0.5	-0.25	-0.25	-0.25

G.4 Hardware

All experiments use an Intel Core i9-7960X CPU with 32GB RAM and an NVIDIA GeForce RTX 2080 Ti GPU.

G.5 Lie Algebra Normalization vs. Lie Group Normalization

This subsection conducts ablations on comparing BN via Lie algebra against our LieBN.

G.5.1 Formulation

An alternative approach to defining the Gaussian distribution is through the Euclidean Gaussian on the tangent space, such as the Lie algebra [59], [89]. In this formulation, the associated mean and variance are reduced to their Euclidean counterparts in the Lie algebra. Based on this, we design the Lie Algebra Batch Normalization (LieAlgebraBN) as:

- Mapping data into the Lie algebra by the Riemannian logarithm at the neural element Log_E ;
- Applying Euclidean BN over the Lie algebra;
- Mapping back to the Lie group by the Riemannian exponentiation at the neural element Exp_E .

Disadvantages of LieAlgebraBN: However, we argue that LieAlgebraBN might distort the geometry as they solely rely on the tangent space. Nevertheless, we compare LieAlgebraBN and LieBN on the SPD and rotation matrices in the following for completeness.

¹⁰. <https://ravitejav.weebly.com/kbac.html>

¹¹. <https://github.com/zhiwu-huang/LieNet>

¹². <https://github.com/hjf1997/LieNet>

G.5.2 Experiments on the SPD Lie Groups

Following App. G.5.1, we implement LieAlgebraBN on the SPD manifolds under LEM, AIM, and LCM, respectively. The neural elements on the SPD Lie groups w.r.t LEM, LCM, and AIM are the identity matrix I . Besides, the logarithmic (exponential) maps at I under AIM and LEM are identical. Therefore, the LieAlgebraBN under AIM and LEM are identical.

We mainly focus on the standard AIM, LEM, and LCM, where no power deformation is applied. However, we observe that LieAlgebraBN under LCM performs significantly worse on FPHA. To address this, we apply a matrix power of 0.5, as Tab. 7c, to activate the LCM geometry. Following the main paper, we conduct experiments on the SPDNet backbone while keeping all other settings consistent.

Tab. 20 presents the 10-fold average comparison on the FPHA dataset. We can observe that our LieBN is consistently better than LieAlgebraBN across different geometries. Notably, LCM-based LieAlgebraBN even exhibits a negative impact on the FPHA dataset, which may be attributed to the distorted geometry of the tangent space. These findings validate the superiority of our intrinsic LieBN over the LieAlgebraBN.

TABLE 20: Comparison of LieAlgebraBN against LieBN under different SPD invariant metrics.

Dataset	BN Type	None	LieAlgebraBN		LieBN		
			AIM/LEM	LCM	AIM	LEM	LCM
FPHA		85.59±0.72	85.63±0.77	77.37±2.28	89.70±0.51	86.56±0.79	86.33±0.43

G.5.3 Experiments on the Rotation Lie Groups

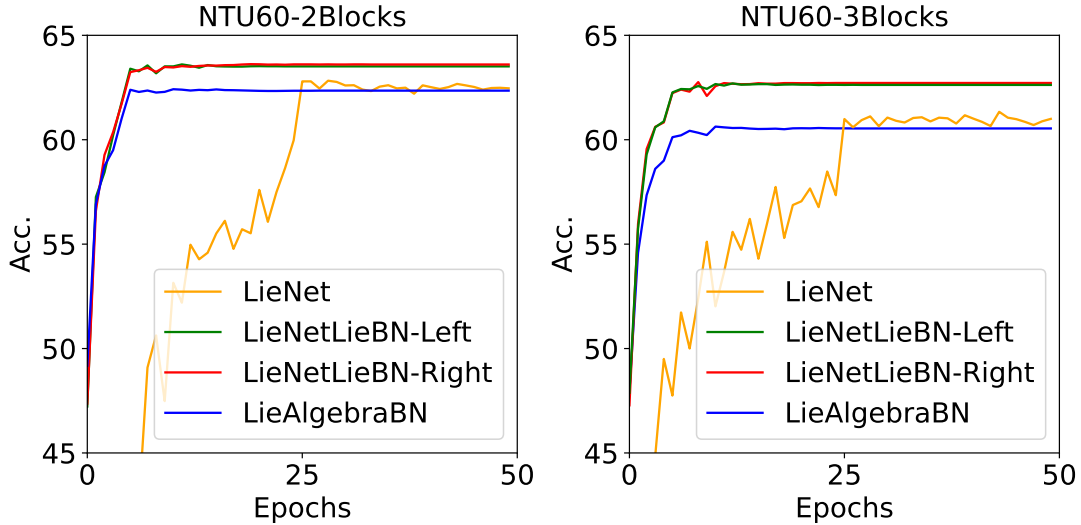


Fig. 5: Comparisons of LieBN and LieAlgebraBN on the LieNet backbone.

We compare LieAlgebraBN and LieBN under the LieNet backbone on the relatively large NTU60 dataset. Following the main paper, we adopt two-block and three-block architectures. We observe that LieAlgebraBN benefits from a decreasing learning rate upon convergence. Therefore, we apply the same learning rate schedule as in LieBN while keeping all other settings identical. Fig. 5 presents the testing accuracy curve, demonstrating that LieBN outperforms LieAlgebraBN under both architectures. This may be attributed to the fact that the Lie algebra, as a local tangent space approximation, distorts the intrinsic geometry of the Lie group, leading to suboptimal normalization.

APPENDIX H

PROOFS

H.1 Proof of Thm. 2.5

Proof. Invariance: First, we note that all four Lie groups are commutative. Secondly, the construction of all four metrics is similar. It suffices to show the left-invariance of ECM.

We only need to show that $L_Q : \mathcal{C}_{++}^n \rightarrow \mathcal{C}_{++}^n$ for any $Q \in \mathcal{C}_{++}^n$ is a Riemannian isometry. L_Q can be rewritten as

$$L_Q = \Theta^{-1} \circ \tilde{L}_{\Theta(Q)} \circ \Theta, \quad (83)$$

where $\tilde{L}_{\Theta(Q)}$ is the left translation over LT_1^n , which is an isometry over LT_1^n . As Θ^{-1} , $\tilde{L}_{\Theta(Q)}$, and Θ are all isometries, their composition is an isometry as well.

WFM: The WFM in the Euclidean space is reduced to arithmetic weighted average. By the isometry of f , one can directly obtain the results. $\square$

H.2 Proof of Thm. 4.1

Proof. **Property 1:** The MLE of M is

$$\begin{aligned} M_{\text{MLE}} &= \operatorname{argmax} \log(K(v)) - \sum_{i=1}^N \frac{d(P_i, M)^2}{2v^2} \\ &= \operatorname{argmin} \sum_{i=1}^N d(P_i, M)^2. \end{aligned} \quad (84)$$

Property 2: We denote $Y = L_B(X)$, and p_X and p_Y as the density of X and Y , respectively. The density of Y is

$$\begin{aligned} p_Y(Q) &\stackrel{(1)}{=} p_X(L_{B_\odot^{-1}}(Q)) \\ &= k(\sigma) \exp\left(-\frac{d(L_{B_\odot^{-1}}(Q), M)^2}{2\sigma^2}\right) \\ &\stackrel{(2)}{=} k(\sigma) \exp\left(-\frac{d(Q, L_B(M))^2}{2\sigma^2}\right). \end{aligned} \quad (85)$$

The above comes from:

- (1) Thm. 7 in [58];
- (2) The isometry of the left translation.

$\square$

H.3 Proof of Thm. 4.2

Proof. The isometry of L_B can directly obtain the homogeneity of the sample mean. Now let us focus on Eq. (15). We have the following:

$$\begin{aligned} \sum_{i=1}^N w_i d^2(\phi_s(P_i), E) &= \sum_{i=1}^N w_i \|s \operatorname{Log}_E P_i\|_E^2 \\ &= s^2 \sum_{i=1}^N w_i \|\operatorname{Log}_E P_i\|_E^2 \\ &= s^2 \sum_{i=1}^N w_i d^2(P_i, E), \end{aligned} \quad (86)$$

where $\|\cdot\|_E$ is the norm on $T_E \mathcal{M}$.

$\square$

H.4 Proof of Thm. 4.5

Proof. As the right-invariant metric shares similar the left-invariant one, this proof follows the similar logic in the above two proofs.

Gaussian homogeneity: We denote $Y = R_B(X)$, and p_X and p_Y as the density of X and Y , respectively. The density of Y is

$$\begin{aligned} p_Y(Q) &\stackrel{(1)}{=} p_X(R_{B_\odot^{-1}}(Q)) \\ &= k(\sigma) \exp\left(-\frac{d(R_{B_\odot^{-1}}(Q), M)^2}{2\sigma^2}\right) \\ &\stackrel{(2)}{=} k(\sigma) \exp\left(-\frac{d(Q, R_B(M))^2}{2\sigma^2}\right). \end{aligned} \quad (87)$$

The above comes from:

- (1) Thm. 13 in [58];
- (2) The isometry of the right translation.

Sample mean homogeneity: This is a direct corollary of the isometry of right translation.

$\square$

H.5 Proof of Thm. 4.6

Proof. As $\mathbb{R}^n$ is an abelian group and the Euclidean inner product is bi-invariant, we focus on left-translation in the following. The core of this proof lies in the fact that on $\mathbb{R}^n$, (1) the Fréchet mean and variance are reduced to the familiar Euclidean statistics. (2) the calculation of the running mean becomes the weighted arithmetic mean. (3) Eqs. (11) to (13) become Eq. (5); We prove these three points one by one.

As stated by Luo *et al* [35, Prop. G.1 and Cor. G.2], from the view of the product manifold, the element-wise Fréchet mean and variance on $\mathbb{R}^n$ are equivalent to the vector-valued Euclidean variance and mean.

Besides, by similar proof as Luo *et al* [35, Prop. G.1], the weighted Fréchet mean on $\mathbb{R}^n$ is simplified as the weighted arithmetic average. Therefore, on $\mathbb{R}^n$, the calculation of running statistics in our Alg. 1 becomes the familiar moving average.

Thirdly, on $\mathbb{R}^n$, we know that $L_x(y) = x + y$, $\text{Exp}_x v = x + v$, $\text{Log}_x y = y - x$, and the neutral element is 0. Since statistics, as well as the Euclidean BN, are calculated element-wisely, we can safely consider a single element, *i.e.* $\mathbb{R}^n = \mathbb{R}$. For a batch of activations $\{x_{i \dots N} \in \mathbb{R}\}$, where the batch mean and batch variance are denoted as μ_b and v_b^2 , then Eqs. (11) to (13) are rewrote as:

$$L_\beta \left(\text{Exp}_0 \left[\frac{\gamma}{\sqrt{v_b^2 + \epsilon}} \text{Log}_0(L_{-\mu_b}(x_i)) \right] \right) = \gamma \frac{x_i - \mu_b}{\sqrt{v_b^2 + \epsilon}} + \beta. \quad (88)$$

The above equation is the exact core computation of the standard Euclidean BN. $\square$

H.6 Proof of Thm. 5.1

Proof. We first prove the case of (θ, α, β) -LEM, and then proceed to the case of θ -LCM.

(θ, α, β) -LEM: For clarity, we denote the metric tensor of (θ, α, β) -LEM as

$$g^{(\theta, \alpha, \beta)\text{-LE}} = \frac{1}{\theta^2} P_\theta^* g^{(\alpha, \beta)\text{-LE}}, \quad (89)$$

where $g^{(\alpha, \beta)\text{-LE}}$ is the metric tensor of (α, β) -LEM. Let $P \in \mathcal{S}_{++}^n$ and $V, W \in T_P \mathcal{S}_{++}^n$, then we have

$$\begin{aligned} g_P^{(\theta, \alpha, \beta)\text{-LE}}(V, W) &= \frac{1}{\theta^2} g_{P_\theta(P)}^{(\alpha, \beta)\text{-LE}}(P_{\theta*, P}(V), P_{\theta*, P}(W)) \\ &= \frac{1}{\theta^2} \langle (\log \circ P_\theta)_{*, P}(V), (\log \circ P_\theta)_{*, P}(W) \rangle^{(\alpha, \beta)} \\ &= \langle \log_{*, P}(V), \log_{*, P}(W) \rangle^{(\alpha, \beta)} \\ &= g_P^{(\alpha, \beta)\text{-LE}}(V, W). \end{aligned} \quad (90)$$

θ -LCM: Let us first review a well-known fact of deformed metrics [90]. Let $\tilde{g} = \frac{1}{\theta^2} P_\theta^* g$ be the power-deformed metric on SPD. Then when θ tends to 0, for all $P \in \mathcal{S}_{++}^n$ and all $V \in T_P \mathcal{S}_{++}^n$, we have

$$\tilde{g}_P(V, V) \rightarrow g_I(\log_{*, P}(V), \log_{*, P}(V)). \quad (91)$$

By Eq. (91), we can readily obtain the results. $\square$

H.7 Proof of Thm. 5.2

Proof. (α, β) -AIM is left-invariant [3]. As the pullback of (α, β) -AIM, (θ, α, β) -AIM is left-invariant as well. Besides, Chen *et al* [69] shows that LCM is the pullback metric from the Euclidean space of LT^n . Therefore, θ -LCM is bi-invariant. $\square$

H.8 Proof of Thm. 5.3

Proof. In the following, we denote the d^L , Log^L , Exp^L as the Riemannian operators under the left-invariant metric, *i.e.* AIM. Note that the Cholesky decomposition pulls back the group operation of matrix product from the Cholesky manifold LT_{++}^n [3]. For simplicity, we abbreviate $\odot^{\text{AI}}$ as $\odot$.

Let us first review the differential map of Cholesky decomposition and its inverse [37, Prop. 4]. Following the notations in this theorem and further denote $X \in T_L \text{LT}_{++}^n$, we have the following

$$\text{Chol}_{*, P}(V) = L \left(L^{-1} V L^{-\top} \right)^{\frac{1}{2}}, \quad (92)$$

$$(\text{Chol}^{-1})_{*, L}(X) = L X^\top + X L^\top. \quad (93)$$

Specifically, for the differential map at I , we have

$$\text{Chol}_{*,I}(V) = (V)_{\frac{1}{2}}, \forall V \in T_I \mathcal{S}_{++}^n. \quad (94)$$

$$(\text{Chol}^{-1})_{*,I}(X) = (X)_{\text{Sym}}, \forall X \in T_I \text{LT}_{++}^n. \quad (95)$$

Denoting $\tilde{L}$ and $\tilde{R}$ as the group translation on the Cholesky manifold LT_{++}^n , we have the following w.r.t. the differential maps of left and right translation:

$$\begin{aligned} \left(L_{P,*,P_{\odot}^{-1}}\right)^{-1} &\stackrel{(1)}{=} \left(\left(\text{Chol}^{-1}\right)_{*,I} \tilde{L}_{L^*,L^{-1}} \circ \text{Chol}_{*,P_{\odot}^{-1}}\right)^{-1} \\ &= \left(\text{Chol}_{*,P_{\odot}^{-1}}\right)^{-1} \left(\tilde{L}_{L^*,L^{-1}}\right)^{-1} \circ \text{Chol}_{*,I}, \end{aligned} \quad (96)$$

$$\text{R}_{P_{\odot}^{-1},*,P} \stackrel{(2)}{=} \left(\text{Chol}^{-1}\right)_{*,I} \circ \tilde{R}_{L^{-1},L} \circ \text{Chol}_{*,P}. \quad (97)$$

The above derivation comes from the following:

- (1) $L_P = \text{Chol}^{-1} \circ \tilde{L}_L \circ \text{Chol}$;
- (2) $\text{R}_{P_{\odot}^{-1}} = \text{Chol}^{-1} \circ \tilde{R}_{L^{-1}} \circ \text{Chol}$.

Riemannian metric: For the differential of right translation, we have the following

$$\begin{aligned} \text{R}_{P_{\odot}^{-1},*,P}(V) &= \left(\text{Chol}^{-1}\right)_{*,I} \circ \tilde{R}_{L^{-1},L} \circ \text{Chol}_{*,P}(V) \\ &= \left(L(L^{-1}VL^{-\top})_{\frac{1}{2}}L^{-1}\right)_{\text{Sym}}. \end{aligned} \quad (98)$$

By Eq. (98), one can obtain the expression for the Riemannian metric tensor.

Riemannian geodesic and exponential map: According to [91], we have the following for the operators between left- and right-invariant metrics:

$$\text{Exp}_P(V) = \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(L_{P,*,P_{\odot}^{-1}}\right)^{-1} \circ \text{R}_{P_{\odot}^{-1},*,P}(V) \right) \right\}_{\odot}^{-1}, \quad (99)$$

$$\text{d}(P, Q) = \text{d}^L(P_{\odot}^{-1}, Q_{\odot}^{-1}). \quad (100)$$

Putting the AIM-based geodesic distance into the RHS of Eq. (100), one can obtain the geodesic distance under CRIM.

Now, we simplify Eq. (99). Putting Eqs. (96) and (97) into Eq. (99), we have the following:

$$\begin{aligned} \text{Exp}_P(V) &= \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(\text{Chol}_{*,P_{\odot}^{-1}}\right)^{-1} \circ \left(\tilde{L}_{L^*,L^{-1}}\right)^{-1} \circ \tilde{R}_{L^{-1},L} \circ \text{Chol}_{*,P}(V) \right) \right\}_{\odot}^{-1} \\ &= \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(\text{Chol}_{*,P_{\odot}^{-1}}\right)^{-1} [L^{-1} \text{Chol}_{*,P}(V) L^{-1}] \right) \right\}_{\odot}^{-1} \\ &\stackrel{(1)}{=} \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(\text{Chol}_{*,P_{\odot}^{-1}}\right)^{-1} \left[\left(L^{-1}VL^{-\top}\right)_{\frac{1}{2}} L^{-1} \right] \right) \right\}_{\odot}^{-1} \\ &\stackrel{(2)}{=} \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(\left(L^{-1}VL^{-\top}\right)_{\frac{1}{2}} L^{-1} L^{-\top} \right)_{\text{Sym}} \right) \right\}_{\odot}^{-1}. \end{aligned} \quad (101)$$

The above comes from the following:

- (1) Eq. (92);
- (2) Eq. (93).

Riemannian logarithm: From the second equality in Eq. (101), we have the following

$$\begin{aligned} \text{Log}_P(Q) &= -\text{Chol}_{*,P}^{-1} \left\{ L \text{Chol}_{*,P_{\odot}^{-1}} \left(\text{Log}_{P_{\odot}^{-1}}^L(Q_{\odot}^{-1}) \right) L \right\} \\ &\stackrel{(1)}{=} -\text{Chol}_{*,P}^{-1} \left\{ \left(L \tilde{V} L^{\top} \right)_{\frac{1}{2}} L \right\} \\ &= - \left(LL^{\top} \left(L \tilde{V} L^{\top} \right)_{\frac{1}{2}} \right)_{\text{Sym}} \end{aligned} \quad (102)$$

The above comes from the following:

- (1) $\text{Chol}_{*,P_{\odot}^{-1}}(V) = L^{-1} (LV L^{\top})_{\frac{1}{2}}, \forall V \in T_{P_{\odot}^{-1}} \mathcal{S}_{++}^n$.

□

H.9 Proof of Thm. 5.4

Proof. Completeness: Eq. (22) indicates that Exp_I is defined over the whole $T_I \mathcal{S}_{++}^n$. Besides, the SPD manifold is connected [50]. By Cor. 6.20 in [43], CRIM is complete.

Geodesic: For simplicity, we abbreviated $\odot^{\text{AI}}$ as $\odot$. The geodesic connecting P and Q can be obtained by the following:

$$\begin{aligned} \gamma(t; P, Q) &= \text{Exp}_P(t \text{Log}_P(Q)) \\ &\stackrel{(1)}{=} \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(- \left(\text{Chol}_{*, P_{\odot}^{-1}} \right)^{-1} [L^{-1} \text{Chol}_{*, P} (t \text{Log}_P(Q)) L^{-1}] \right) \right\}_{\odot}^{-1} \\ &\stackrel{(2)}{=} \left\{ \text{Exp}_{P_{\odot}^{-1}}^L \left(t \text{Log}_{P_{\odot}^{-1}}^L (Q_{\odot}^{-1}) \right) \right\}_{\odot}^{-1} \\ &= \left\{ \gamma^{\text{AI}}(t; \tilde{P}, \tilde{Q}) \right\}_{\odot}^{-1}. \end{aligned} \quad (103)$$

The above comes from the following:

- (1) The second equality in Eq. (101);
- (2) The first equality in Eq. (102).

□

H.10 Proof of Thm. 5.6

Proof. With out loss of generality, we focus on the case of the left-invariant metric. The results of the right-invariant metric can be proven similarly.

We denote Eqs. (11) to (13) on $\mathcal{M}_i, i = 1, 2$ as the mapping $\xi^i(\cdot | M, v^2, B, s)$. Let $\mathcal{B} = \{P_{1\dots N}\}$ and $f(\mathcal{B}) = \{f(P_{1\dots N})\}$.

The core of this proof lies in three points:

- 1) The Fréchet mean and variance of $\mathcal{B}$ in $\mathcal{M}_1$ correspond to the counterparts of $f(\mathcal{B})$ in $\mathcal{M}_2$.
- 2) $\xi^1(P_i | M, v^2, B, s)$ in $\mathcal{M}_1$ is equal to $f^{-1}(\xi^2(f(P_i) | f(M), v^2, f(B), s))$.
- 3) The updates of running statistics in $\mathcal{M}_1$ correspond to the counterparts in $\mathcal{M}_2$.

We denote M as the Fréchet mean of $\mathcal{B}$, and v^2 as the Fréchet variance of $\mathcal{B}$. Then, by the isometry of f , the Fréchet mean and variance of $f(\mathcal{B})$ are $f(M)$ and v^2 , respectively.

On $\mathcal{M}_i, i = 1, 2$, we denote $L^i, \odot^i, \text{Exp}^i, \text{Log}^i$ as the Lie group and Riemannian operators, E^i as the neutral element, and Eq. (12) as $\phi_s^i(\cdot)$. With the isometry and Lie group isomorphism of f , we have the following equations:

$$L_{M_{\odot}^{-1}}^1 = f^{-1} \circ L_{f(M)_{\odot}^{-1}}^2 \circ f, \quad (104)$$

$$\begin{aligned} \phi_s^1 &= \text{Exp}_{E^1}^1 \left[s \text{Log}_{E^1}^1(\cdot) \right] \\ &= f^{-1} \left(\text{Exp}_{E^2}^1 \left[s \text{Log}_{E^2}^2(f(\cdot)) \right] \right) \\ &= f^{-1} \circ \phi_s^2 \circ f, \end{aligned} \quad (105)$$

$$L_B^1 = f^{-1} \circ L_{f(B)}^2 \circ f. \quad (106)$$

Then we have

$$\xi^1(P_i | M, v^2, B, s) = f^{-1}(\xi^2(f(P_i) | f(M), v^2, f(B), s)) \quad (107)$$

Lastly, we show the correspondence between running statistics. Since the Fréchet variance is the same for both $\mathcal{B}$ and $f(\mathcal{B})$, we focus on the running mean. Let M_r and $f(M_r)$ denote the initial values of the running means in $\mathcal{M}_1$ and $\mathcal{M}_2$ respectively, and WFM^i represent the weighted Fréchet mean in $\mathcal{M}_i$. Then the updated running mean in $\mathcal{M}_1$ is

$$\text{WFM}^1(\{1 - \gamma, \gamma\}, \{M_r, M\}) = f^{-1}(\text{WFM}^2(\{1 - \gamma, \gamma\}, \{f(M_r), f(M)\})) \quad (108)$$

we can further simplify the above equation as

$$\text{WFM}^1 = f^{-1} \circ \text{WFM}^2 \circ f \quad (109)$$

Denoting LieBN^i as the LieBN algorithm on $\mathcal{M}_i$, Eq. (107) and Eq. (109) imply that:

$$\text{LieBN}^1(P_i | B, s, \epsilon, \gamma) = f^{-1} \left[\text{LieBN}^2(f(P_i) | f(B), s, \epsilon, \gamma) \right]. \quad (110)$$

□